



Teamcenter 13.3

Manufacturing Execution System Integration

Teamcenter 13.3

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1. Getting started with the Manufacturing Execution System Integration

Overview of the Manufacturing Execution System Integration

A manufacturing execution system (MES) is designed to help companies more effectively and efficiently execute manufacturing operations from the product order through each step of the manufacturing process to its final point of delivery. Equally important, an MES is a dynamic information system that is key to collaborative manufacturing strategies by providing mission-critical information about production activities to managers across an organization and its supply chain.

The integration between the definition process in product lifecycle management (PLM) and the control and execution process in an MES generates tremendous value for both processes. As part of the PLM process, the bills of process (BOPs) can be fully simulated in digital replications of the factory. These BOPs can be created, tuned, potentially optimized, validated, and then sent as a technical work package to an MES that automatically generates a work plan for the shopfloor. The MES also controls and monitors the execution of this work plan on the factory floor. Therefore, the interoperability between the MES and PLM is critical in closing the gap between simulation in the engineering environment and reality on the factory floor. As production ramps up, information collected by the MES is essential in adjusting the bills of process. This information is fed back to engineering to improve their knowledge of the actual BOPs needed to produce desired results. Continuous improvement initiatives can benefit from the collection of data when the MES is monitoring and reporting what actually occurred on the physical factory floor.

The Manufacturing Execution System Integration collects the bill of process, the bill of materials, and any relevant work instructions into a work package that is released to the MES. The operator can use this information to machine the desired product. If there is a problem with the data, the planner can modify the contents of the work package and release it again.

Before you begin

Prerequisites	MES Integration requires a separate license in addition to the standard Teamcenter product license. You must obtain and install the appropriate license required to run MES Integration.
Configure MES Integration	• Install the software. • Configure the integration to meet your business needs (for example, define attributes in Teamcenter and specify on which form types they are stored). • Set preferences and global settings. • Define a mapping schema in the mapping file.

- Set up workflow procedures to handle validation and data transfer to MES.

Start MES
Integration

There is no dedicated interface to integrate with MES. To perform tasks necessary to send data to MES, choose **Tools→MES** in the Manufacturing Process Planner application.

MES Integration interface

The MES Integration has no dedicated interface within Teamcenter. The following Teamcenter applications and components are used in the integration.

Application/component	Use
Manufacturing Process Planner application	• Create the execution plan. • Create work instructions. • Run various MES-related commands using the Tools→MES menu commands and the shortcut menu in the Collaboration Context Tree view or the process structure view. • View MES data in the Web Browser.
Workflow Designer application	Modify the default workflow used to collect the work instructions, bill of materials, and bill of process into a work package.
Edit→Options menu command	Set all the preferences necessary to configure the integration.
Application interface viewer	Monitor the transfer of the work package to the MES.
My Teamcenter application	Search for the work package using the MES Work Package query. Open the work package in the application interface viewer by right-clicking the work package and choosing Open with→Viewer .
PLM XML/TC XML Export Import Administration application	Configure the transfer rules that you use to specify the contents of the work package.

Basic concepts for using the MES Integration

You encounter the following concepts and terms when working with this integration.

Term	Description
Product definition (MBOM, EBOM)	A definition of the product and its requirements for manufacturing in the proper granular level required by manufacturing.
Bill of material	A summarized list of materials that are consumed.
Master execution plan	A collaboration context containing a process plan that describes how a product should be manufactured. The master plan is an execution plan that is relevant to the base product without additions or changes for a specific order. The master execution plan contains the product and optionally the plant definition. It can have a finite configuration (although specific orders can have some deviations in the process), or, in some cases, it may correspond to multiple products (that is, 150% or 120% BOM). Also referred to as a <i>master</i>.
Order-specific execution plan	An execution plan that is derived from a master execution plan for a specific order (date, effectivity) for a specific work area. An order-specific execution plan is for one time execution by a manufacturing execution system. Also referred to as an <i>order</i>.
Operation	A behavior description of a manufacturing process that is managed and monitored.
Compound operation	A dedicated process type that holds the execution steps in an execution plan.
Execution Step	The implementation of an operation by distinct working instruction blocks; each such block is an execution step. The completion of an execution step is typically logged and monitored against the as-built repository, so that the completion of tasks can be tracked.
Product production rule (PPR)	The manufacturing execution system steps required to make a product. A product production rule (PPR) represents the process plan or bill of process for a product.
Production order (or work order)	A request to build a product for which a process plan exists. It usually originates with an ERP system.
Work package	The packaging of the execution plan for transferal to the MES.
Engineering nonconformance/issue	A situation where an operator fails to execute an instruction (or encounters a discrepancy between the execution and the plan) and requires planner intervention to determine

Term	Description
	what to do next. Such discrepancies require mechanisms to report back from execution to planners and engineers to request their involvement. The planner's resolution of the issue can require an update to execution plans (and subsequently to the work package). Creating such an operator's report is called <i>red-lining</i> .
Application Interface object	A persistent workspace object that is the repository for the export transactions from Teamcenter to the external MES application. It contains the work package to be transferred to the MES application. Application interface objects are listed in My Teamcenter and you can view them in the viewer in that application. This viewer is referred to as the application interface viewer and provides a list of all application interfaces awaiting transfer to the MES.
Transfer mode	A persistent workspace object that contains instructions for Teamcenter on which data to collect for a specific purpose.

Basic tasks for using the MES integration

1. Create a master execution plan containing all the data to send to the MES.
2. Validate the data for the execution plan.
3. Work through the validation results, solving any problems found by the validation.
4. Send the order execution plan to the MES.
5. Respond to any nonconformance issues raised by the operator.
6. Send a new package to the MES that contains only the changes made to the previous plan.

2. Installing the MES Integration

Installing the MES Integration

- Install the work instructions template on top of Teamcenter by running Teamcenter Environment Manager (TEM) and selecting the **Extensions→Manufacturing Process Management→Work Instructions** feature.
- Install one of the following integrations:
 - Install the MES integration in TEM by selecting the **Extensions→Manufacturing Process Management→MES Integration** feature; OR
 - Install the closed loop manufacturing (CLM) integration in TEM by selecting the **Extensions→Closed Loop Integration** feature.
- (Optional) To release a plant BOP structure to production by using change objects, install Change Management in TEM by selecting the **Extensions→Enterprise Knowledge Foundation→Change Management** feature.
- After upgrading to the next version of Teamcenter, refer the `TC_ROOT\install\cmtmesinteg\Help_Patching_cmtmesinteg_Solution.txt` file to upgrade MES Integration.
- To create work instructions with standard text, ensure that Office 2007 or later is installed.
- To create 2D snapshots, ensure that the **Image Capture** option is selected in the Teamcenter **Options** dialog box.
- To manage nonconformance issues, ensure that the **MES Issue Management** feature is installed in TEM.
- To generate work instruction representations, ensure that Dispatcher, including the asynchronous service and **MesDataGenerationTrans** translator, is installed.
- Set the required preferences.

Understanding the ReleaseToMES workflow

The integration between Teamcenter and the MES is based on workflows. In Teamcenter, a workflow captures business procedures where documents, information, or tasks are passed from one participant to another. The MES Integration includes a workflow that validates the data, gathers the required documents and structures, and prepares them for transfer. The **ReleaseToMES** workflow has two input parameters: the execution plan, a collaboration context of type **ExecutionPlan**, and the application interface object of type **Execution Plan Application Interface**.

The **ReleaseToMES** workflow performs two main actions:

- **Validates data**
Before releasing data to the MES, the **ReleaseToMES** workflow checks the data to ensure it is compatible with the MES data model. It runs through all the registered validations. This check uses the same workflow handlers that validate the data when you use the **Validate for MES** menu command.
- **Transfers data**
The workflow ensures that the metadata and data are ready to be pulled from the MES side. To do this, it creates one application interface request per transfer that represents all required data and metadata.

Understanding application interfaces

An application interface is a persistent workspace object that is the repository for the export transactions from Teamcenter to the external MES application. It contains the work package to be transferred to the MES application.

Application interface objects are listed in My Teamcenter, and you can view them in the application interface viewer in that application. This viewer provides a list of all application interfaces awaiting transfer to the MES.

Understanding transfer modes

A transfer mode is a persistent workspace object that contains instructions for Teamcenter on which data to collect for a specific purpose. Transfer mode objects are used throughout Teamcenter. They allow users to import, export, or exchange data with little knowledge other than the name of the transfer mode object that they should use. For example, **ToCompanyA**, **FromCompanyB** or **ToCompanyB**, are transfer modes that are created in PLM XML/TC XML Export Import Administration and are displayed as options when you import or export objects or system data using one of the Teamcenter applications.

In the context of the MES Integration, the transfer mode defines what data is included in the work package that is transferred to the MES. Transfer mode objects are made up of the following items that configure the import or export operation.

- **Closure rules**
Define the scope of the data translation.
- **Filter rules**
Use conditions to apply a finer level of control over which data gets translated along with the primary objects.
- **Action rules**

Specify sets of methods that can be called before, during, and after the translation.

- **Property sets**

Provide a mechanism for PLM XML objects to have arbitrary properties in the form of **UserData** elements.

Configuring the ReleaseToMES workflow

You can configure the MES workflow to suit your needs. For example, you can add a task at the end of the workflow that sends an e-mail to the initiator informing that the workflow is complete.

The **Review** action handler is not part of the default workflow. You can add it to your workflow but be aware that the last reviewer to approve initiates the continuation of the workflow on that machine. This can potentially overtax the last reviewer's computer until the workflow is complete. To work around this:

- Create a separate workflow to manage the review task.
- Add the workflow initiator as a reviewer to the review task so that the reviewer is always the last person to approve and the workflow is initiated on that machine.
- Run the workflow asynchronously. This passes the task to Dispatcher to run at an appointed time.

Understanding validity checks

You can check that all data meets specific requirements by creating a validity check for the data. By default, several validity checks are included with integration. These are run when you choose the **Validate for MES**, **Release to Execution**, or **Release to Production** menu commands.

Flow Validation	Verifies that there is a full flow between the compound operations and steps. It verifies there is only one beginning and one end to the flow. In the case of compound operations or operations, you can have parallel flows.
Process Hierarchy Validation	Verifies that the structure of a process is valid. A valid process structure adheres to the following: • The root of the process structure is a Process object. • There is a maximum of two levels of processes. • The process has at least one child compound operation in the assembly structure. • The process has at least one child operation in the part machining structure. • There are no multilevel compound operations.

- The process does not contain a mix of compound operations, operations, and subprocesses.

The validation is performed from the top down. If the process object is validated, the validation runs on the entire structure. If the validation is performed on a compound operation, only the selected compound operation and its children are validated.

Release Status Validation	Verifies that all data types and subtypes have one of the release statuses defined in the MEMESReleaseStatus preference.
Workarea assigned validation	Verifies that every item in the process structure that is of the type specified in the MEMESWorkAreaAssignedValidationObjects preference has one work area item assigned. By default, only the compound operation type is defined in this preference.
Workarea name validation	Verifies that the name of the work area assigned to the process structure does not contain any spaces.

Register validity checks

You can check that all data meets specific requirements by creating a validity check for the data. By default, several validity checks are included with integration.

You can configure additional validity checks to suit your business needs.

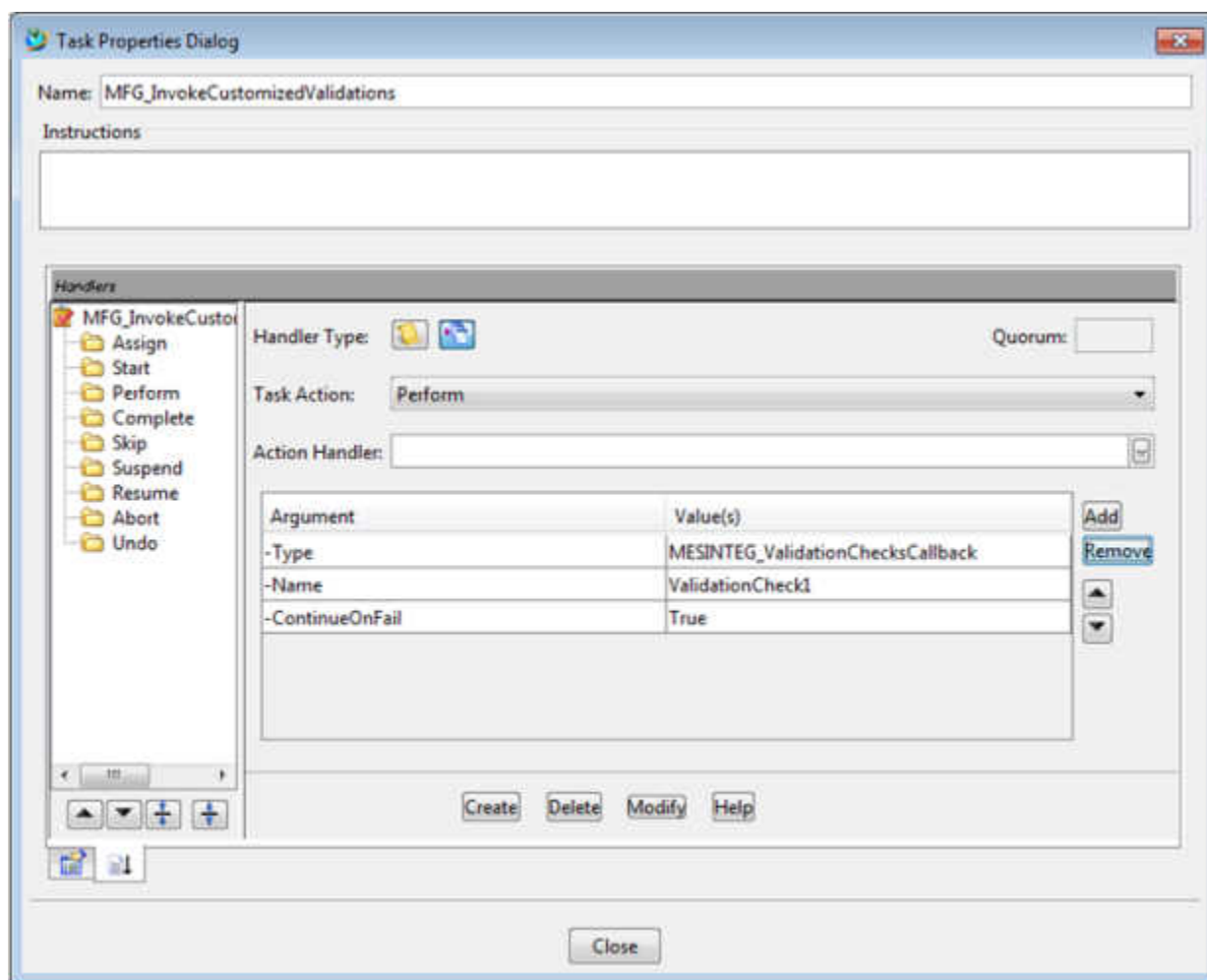
Before releasing data to the MES, you can call the **Validate for MES** menu command to run the validity checks. In addition, the **ReleaseToMES** workflow checks that the data is ready to be released to the MES. These checks are run by the **MFG-invoke-customized-validations** rule handler that is part of the workflow.

You can write custom validation checks and add them to the workflow.

1. Write a custom validation check that validates the required information.
2. Register the validation check as a callback function by running the **install_callback** utility.

```
install_callback [-h] -u=user -p=password -g=group
                 -mode=mode {install | create | modify | delete | list }
                 -type=MFG_ValidationChecksCallback
                 [-library=name-off-dll-containing-the-validity-check]
                 [-function=C++-function-name-of-the-registered-validity-check]
                 -name=callback-name
```

3. In Workflow Designer, add the validity check to the **MFG-invoke-customized-validations** rule handler. In the example, a validity check with the name **ValidationCheck1** is added to the rule handler.



The combination of type and name must be unique. These two entries are treated as a pair. You can register multiple callback functions by listing them in the **name** argument, separated by commas. For each name, you must list a corresponding type.

-Type= MFG_ValidationChecksCallback,MFG_ValidationChecksCallback,
MFG_ValidationChecksCallback

-Name= ValidationCheck1, ValidationCheck2, ValidationCheck3

If you are using multiple validity checks, setting the **ContinueOnFail** argument to **True** causes Teamcenter to continue on to the next validity check even if the previous one fails.

3. Authoring the data

Objects included in an execution plan

The data that is transferred to an MES is contained in an *execution plan*. An execution plan is a dedicated collaboration context of type **Execution Plan**. The execution plan consists of a process structure context containing a root process.

Within the root process, the execution plan consists of multiple processes. Each process represents a mini-bill of process (mini-BOP) that is the process required to manufacture a subassembly of a product. The collection of mini-BOPs inside the execution plan is the amount of work sent to the MES as a unit or package.

The subassembly being manufacturing by the mini-BOP must be associated to the mini-BOP using the **Associate Product as Target** menu command. This links the two objects with a **Mfg. Targets** relation.

An execution plan can contain the following objects:

- Compound operations
- Execution steps under compound operations
- Parts consumed by execution steps
- Tools assigned to execution steps
- Work areas allocated to compound operations
- Attached data such as textual work instruction documents, product views, snapshots, or other documents

Note:

Structure Manager allows you to copy and paste execution steps from a compound operation to another compound operation.

Setting up the data

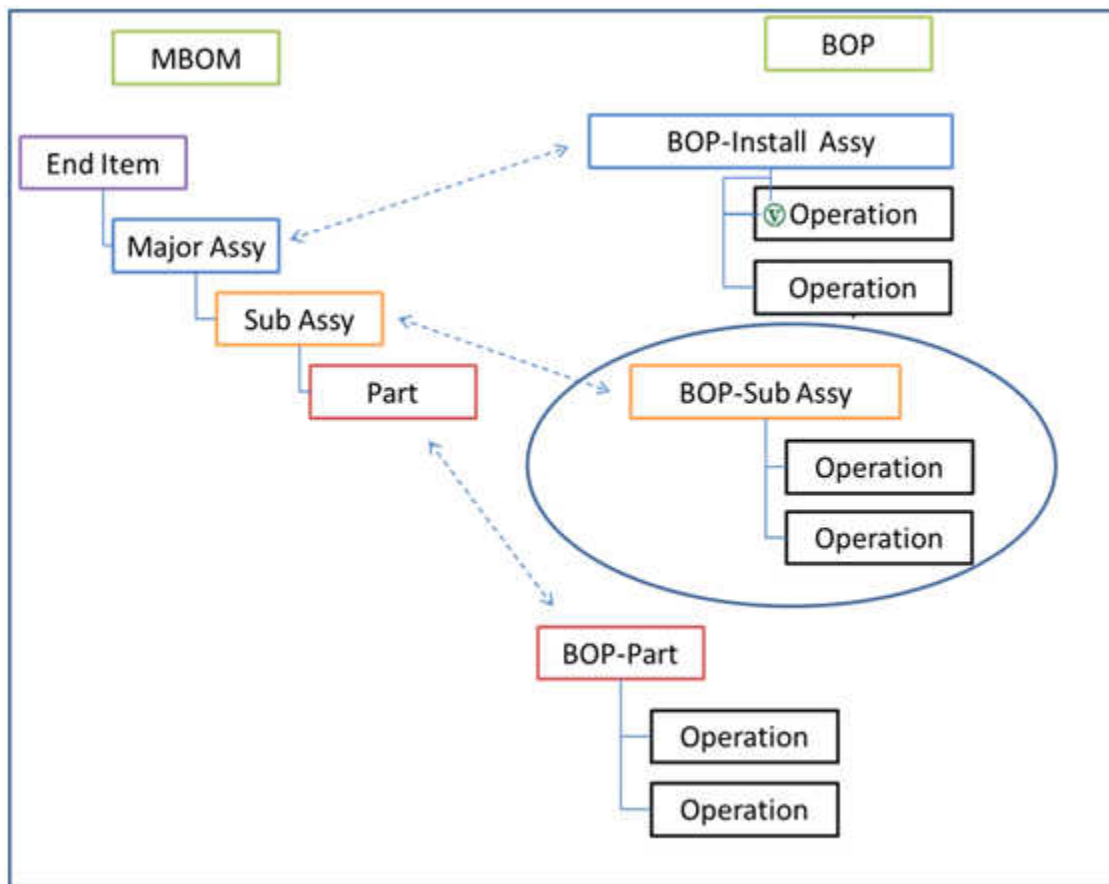
There are several assumptions when working with the integration.

- The MBOM contains business parts (the parts in the MBOM should include part numbers or catalog numbers).
- The plan is plant-aware.

- A work area is allocated to an operation.
- The name of a work area is not changed when revising.
- The work area name is unique in the factory structure.

Modeling for manufacturing versus modeling for assembly

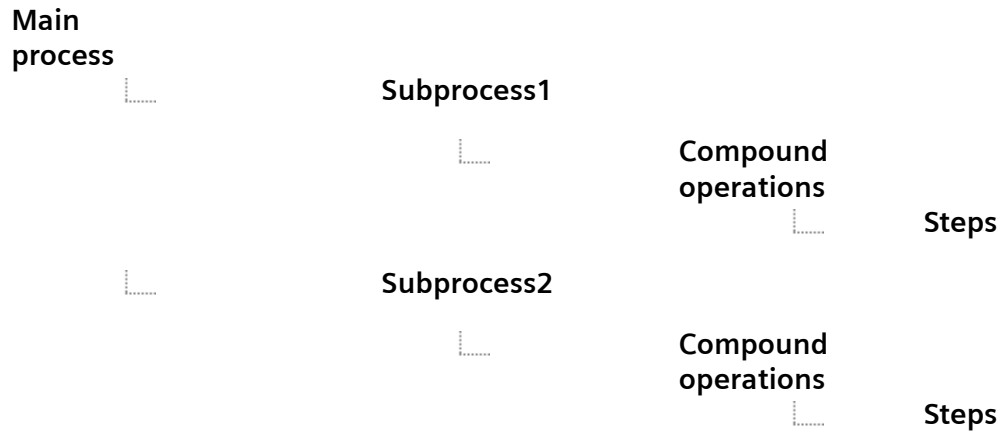
In general, you should have an MBOM as a separate structure under the execution plan (see the diagram for an example) from which parts are consumed into steps/operations. The structures of the MBOM and BOP are not necessarily parallel.



From the plant structure, work areas in the plant are allocated to operations in the BOP. Tools are used by steps/operations.

For assembly planning:

The bill of process for assembly planning should contain a list of processes (that represent product production rules) under a parent process.



When sending this structure to the MES, two product production rules (PPRs), **Subprocess1** and **Subprocess2**, are created. The parent PPR can be created manually in the MES or can come from an ERP system.

The process represents the root of the mini-BOP to manufacture the part or subassembly.

- The process type to be used by default is **Process**. You can use other process types, but you must configure the integration mapping file.
- The default operation type to use is **Compound Operation** (derived from **Process**). You can use other process types, but you must configure the integration mapping file. You create the compound operation by clicking  in the view toolbar.
- You create the **ExecutionStep** object under the operation by clicking  in the view toolbar. The execution step is derived from the EOP class.
- The operation or execution step has the items and tools under it that are consumed during the operation.

For part manufacturing:



The process represents the root of the mini-BOP to manufacture the part.

- The process type to be used by default is **Process**. You can use other process types, but you must configure the integration.
- The default operation type to use is **OP**. You can use other process types, but you must configure the integration for other process types.

- This operation type should be part of the **Operation** class (**File** → **New** → **Operation**).
- You create activities under the operation using the **File** → **New** → **Activity** menu command or they can come from the CAM system.
- The operation has the items and tools under it that are consumed during the operation.

Differentiating between a master execution plan and an order

When setting up the data to send to the MES, you can create an order-specific execution plan. You specify the order in Manufacturing Process Planner using configuration based on unit effectivity. The unit used as unit effectivity is mapped to a specific and unique order ID. You must define and remember the correlation between the unit effectivity you use in the configuration rule and the order ID that the execution plan represents. If the order ID does not contain any nonnumeric characters, you can use the order ID as the unit number. When authoring objects for the master plan, you set the effectivity on this object to be effective for all orders. To achieve this, you set the effectivity range as **0-UP** on the process lines. When configuring for a master or an order, both the master and all the orders derived from it reflect the changes.

You must set the configuration rule manually. Teamcenter does, however, provide a dedicated command to add the unit effectivity to the execution plan.

Note:

Manually setting the configuration means that you must make a choice between repeatedly editing the revision rule or creating a large number of execution plans.

Teamcenter automatically assigns the correct effectivity to any object revised or created within an execution plan. It uses workflows to differentiate between objects that are already released and objects that are changed and therefore must be included in an incremental package.

Create an execution plan

The object that is sent to the MES is an *execution plan*. An execution plan is a collaboration context of the type **Execution Plan**. The execution plan consists of a process, its compound operations, and execution steps as well as any consumed objects such as parts, resources, or work areas. Teamcenter automatically sends a notification when a changed master execution plan is released.

When creating an execution plan, you have several choices of how to build the structure.

- Top down:
 1. Create an empty collaboration context of type **Execution Plan**.
 2. Create an empty **Process** structure context within the execution plan.

3. Fill the content of the new structure context by creating objects or copying and pasting existing objects.
- Bottom up:
 1. Open a process whose structure conforms with the requirements (for example, structure, types, level, assignment).
 2. Save the process as an **Process** structure context.
 3. Save all structure contexts in the session to a new execution plan collaboration context or move structure contexts into an existing execution plan.

To create the objects in an execution plan:

Object	Is a	Create by choosing
Execution plan	Collaboration context	File → New → Collaboration Context → Execution plan
Process root	Process	File → New → Process (Process) or click  in the process structure view.
Operation	Operation	File → New → Operation or click  in the process structure view.
Compound operation	Process	Click  in the process structure view.
Execution step	Operation	Click  in the process structure view.

To perform a validation check on an execution plan:

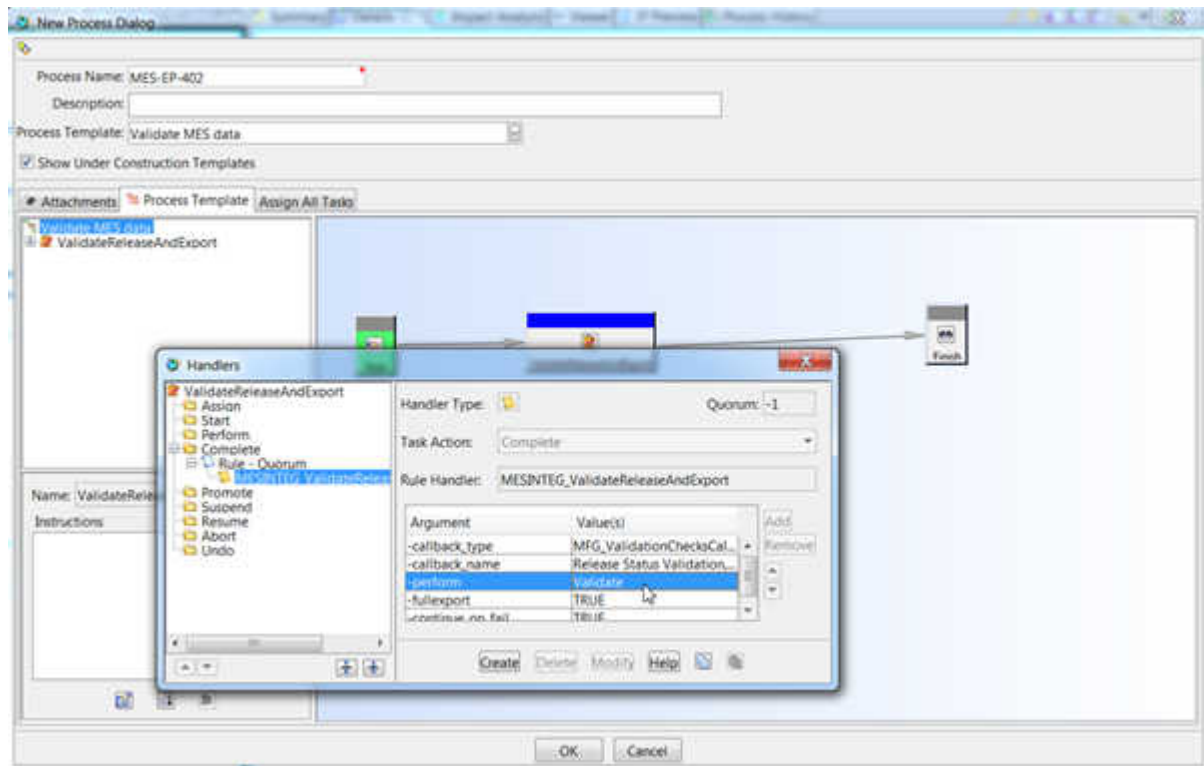
1. Create a workflow template for the **MESINTEG_ValidateReleaseAndExport** rule handler with **Validate** as the value for the **-perform** argument.

The **MESINTEG_ValidateReleaseAndExport** rule handler performs validation checks for MES Integration.

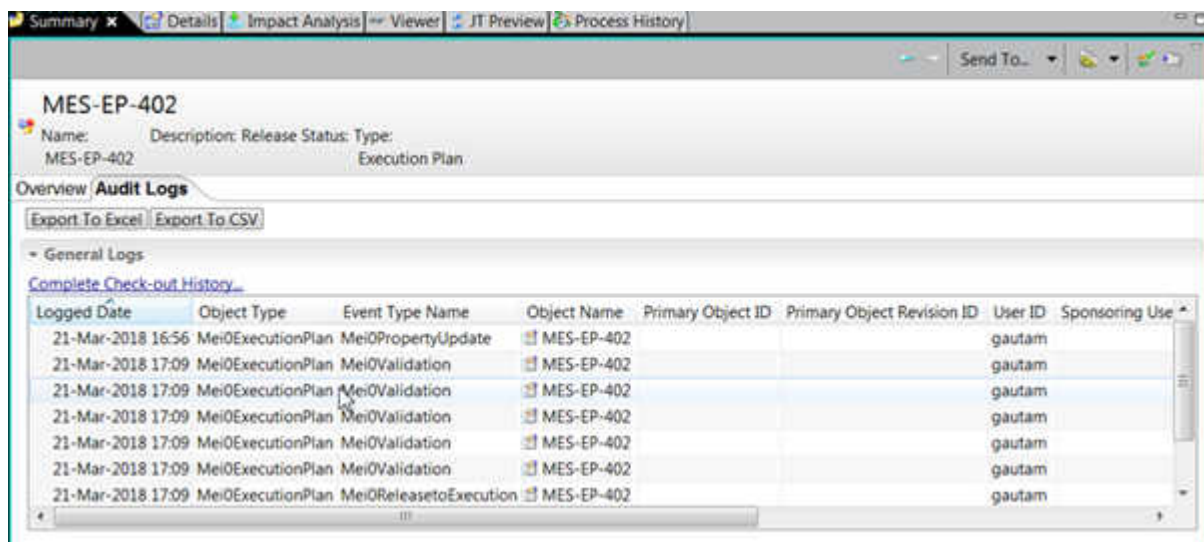
Note:

Release to MES using the **MESINTEG_ValidateReleaseAndExport** rule handler is *not* supported for plant BOPs.

2. In My Teamcenter, select the execution plan and click **File → New → Workflow Process**.
3. In **New Process Dialog**, specify a process name and select the workflow process template you created in step 1 and click **OK**.



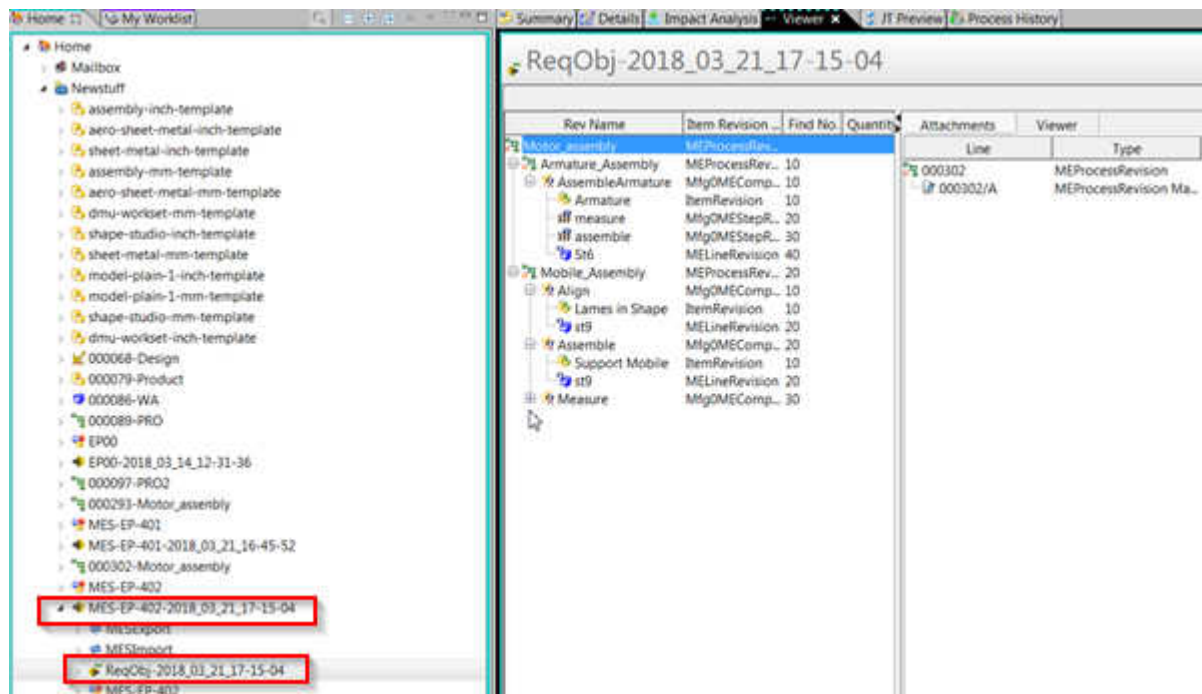
4. To view the validation logs, select the execution plan from the **Home** folder in My Teamcenter, and in the **Summary** tab, click **Audit Logs**.



5. If there are no validation errors, you can use another workflow process to release the execution plan to MES Integration.

After the workflow process is complete, you can refresh the folder in the **Home** view of My Teamcenter. You can expand the execution plan to view the application interface objects and request objects created by the process after performing a release.

The **ReqObj** object contains the structure sent to MES Integration.



6. To view the logs, select the **Summary** tab and click **Audit Logs**.

Provide the authorization to perform signoffs and validations

On the shop floor, only operators with specific skills and the necessary skill certificates can perform certain manufacturing operations. After creating an operation in an execution plan, the manufacturing planner can define role-based signoffs at the completion stage, electronic signature validations, and skill-certification validations for the shop floor.

- Role-based signoffs are performed at the completion stage for a manufacturing operation or for an execution step in order to conform with the appropriate specifications.
- Electronic signature validations are performed for operators to start, pause, or complete a manufacturing operation or an execution step.
- Skill-certificate validation are performed for operators to carry out a manufacturing operation or an execution step. For example, only an operator with specific skills and the necessary skill certificate can perform a particular manufacturing operation.
- The **Buy-Off DCD** type of data collection definition (DCD) is no longer supported under an operation or execution step. However, if there is a **Buy-Off DCD** type (supported in releases prior to 10.1.6) under an operation or execution step, then the SIMATIC IT implementation requires the signoff **Required** option. You can set this by using the **Set Execution Properties** dialog box.

1. Right-click an operation or step and choose **MES→Set Sign-off/E-Signature and Skill....**

Alternatively, select the operator or step and click the **Set Sign-off/E-Signature and Skill...** toolbar button

2. (Optional) To enable signoff, select the **Required** option, and from the **Approver role** list, choose a role.

Note:

This step is intended for a user who is authorized to signoff work performed by shop floor operators.

The default approver role values are **Inspector**, **Certification Engineer**, **Quality Engineer**, and **Stamp Authority**. These values are defined in the **Mei0MESSignOffRoles** list of values (LOV) in Business Modeler IDE.

3. (Optional) To enable an electronic signature, select **Start** at the start of an operation or a step, **Pause** to pause an operation or a step, or **Complete** after completing the operation or the step.

This step is intended for a shop floor operator, for example, an operator who is authorized to log on to the SIMATIC IT operator terminal.

4. (Optional) To enable a skill certificate, select **Required**.

This step is intended for a skilled shop floor operator, for example, an operator who is authorized to log on to the SIMATIC IT operator terminal.

Authoring work instructions

You can create textual work instructions in Teamcenter and associate them with a step in the execution plan.

The work instructions are included as attachments when releasing the data to the MES and can be made available for viewing in the MES.

From the MES side, the textual work instructions can appear as follows (the exact appearance depends on the MES implementation; in this case, it is with SIMATIC IT).



The Teamcenter viewer is on the left, the textual work instructions is in the middle, and an MES form reflecting the DCDs is on the right.

Tracking changes to an execution plan

Teamcenter uses status to track which compound operations and execution steps in an order are ready to be sent to the MES or have already been sent to the MES. By default, it uses the following statuses:

- MES Work In Progress**
 When you start authoring the BOP structure, Teamcenter applies this status to all new lines. The lines within a structure that have only this status are in an initial or working state. You can work on these lines until you are ready to release them.
- MES Pending to Release**
 After you complete the authoring of a BOP line and it is ready to be released to the MES, you must apply this status. You can no longer make changes to this line. You can add this status to multiple lines at once using the **Apply Order Effectivity** command.
- MES Released**
 Once the data is successfully released, Teamcenter applies this status to all released lines in the bill of process.

These statuses are set by workflows that are initiated when you perform various actions.

When you initially work on an execution plan, the lines have an **MES Work In Progress** status. When you are ready to send the plan to the MES, the lines are marked with an **MES Pending to Release** status.

001913/B;1-comp_op (View)	MES Work In Progress 66 (001910)	MES Pending To Release 66 (001910)
001921/B;1-ex_step	MES Work In Progress 66 (001910)	MES Pending To Release 66 (001910)
001922/B;1-add_ex_step	MES Work In Progress 66 (001910)	MES Pending To Release 66 (001910)

When you send an order to the MES the first time, Teamcenter sends all the lines that are marked as **MES Pending to Release** to the MES and marks them as **MES Released**. Once a line is marked as pending or released, you cannot change it until you revise the line.

Revising a released line so that you can work on it causes the line to be marked with the **MES Work In Progress** status again. When you are finished making all changes, you change the status to **MES Pending to Release** again. The next time you send an order, Teamcenter looks to see if the execution plan has already been released to the MES. If so, you can choose to send the same order again, or only send the changes that have happened since the last time you sent the plan. If you choose to send only changes, Teamcenter looks for lines that are revised and marked with the **Pending for MES** release status and highlights them. You can then release these changes. This updates the released work package with a new request with these changes. The modified lines, which are now released, receive an **MES Released** status. Changes are tracked using the incremental change mechanism.

You can view the release status in two ways:

- Add the **bl_rev_release_statuses** or the **Rule configured by** column to the view.
- Open the **Attachments** view on the desired line.

You can change the statuses used to specify the different states in the following preferences:

- **MEMESPendingReleaseStatus**
- **MEMESReleasedStatus** to specify the release statuses attached to the item revisions and incremental change revisions after they are exported to the manufacturing execution system.
- **MEMESWorkInProgressReleaseStatus** to specify the release status to be attached to the item revisions and incremental change revisions on creation.

Setting up the revision rule

You must set up the revision rule correctly to configure the desired revision lines for export. Whether initially releasing a structure or releasing an update, it is important to configure the structure based on the MES statuses.

The configuration is based on unit effectivity. In this integration, a unit number of 0 represents a master bill of process. A nonzero number as unit effectivity represents an order number. The end item for configuration is the top line of the process structure.

The following is a possible revision rule used to configure a structure for authoring order number 200.


```

Has Status( MES Work In Progress, Configured Using Unit Number )
Has Status( MES Pending To Release, Configured Using Unit Number )
Has Status( MES Released, Configured Using Unit Number )
Has Status( Any Release Status, Configured Using Released Date )
Unit No.( 200 )
End Item( 001071-LH-Wing )
Date( Today )

```

Before releasing the structure to the MES, you can change the revision rule as shown to configure **MES Pending** lines.

```

Has Status( MES Pending To Release, Configured Using Unit Number )
Has Status( MES Released, Configured Using Unit Number )
Unit No.( 200 )
End Item( 001071-LH-Wing )
Date( Today )

```

Why use CLM integration?

Closed loop manufacturing (CLM) connects PLM and MES systems by providing a tighter integration with SIMATIC IT user interface, better performance, and lighter data exchange.

Using the CLM integration, you can:

- **Link Teamcenter objects such as work area and resources to SIMATIC IT plant model.**
- **See the work orders associated with the respective Teamcenter execution plan.**
- Convert the Teamcenter manufacturing process structure to the canonical object model structure.
- Send product and manufacturing information (PMI) in the BOP to SIMATIC IT for data collection or inspection.
- Create work instructions in the Easy Plan text editor and view them in SIMATIC IT user interface during work order execution.
- Send Teamcenter datasets using the CLM integration.

When you create a BOP, Teamcenter provides a high level of flexibility to model your data. When you are ready to release the BOP to an execution plan, Teamcenter creates the *As Planned* BOP. The process of transforming the BOP baseline model to the *As Planned* BOP canonical model is called normalization. Normalization is based on default behavior and rules that are derived from customer-specific configuration and data model extensions.

The As Planned BOP is a representation of the Teamcenter BOP in the SIMATIC IT canonical data model. You can use this data in SIMATIC IT to create orders.

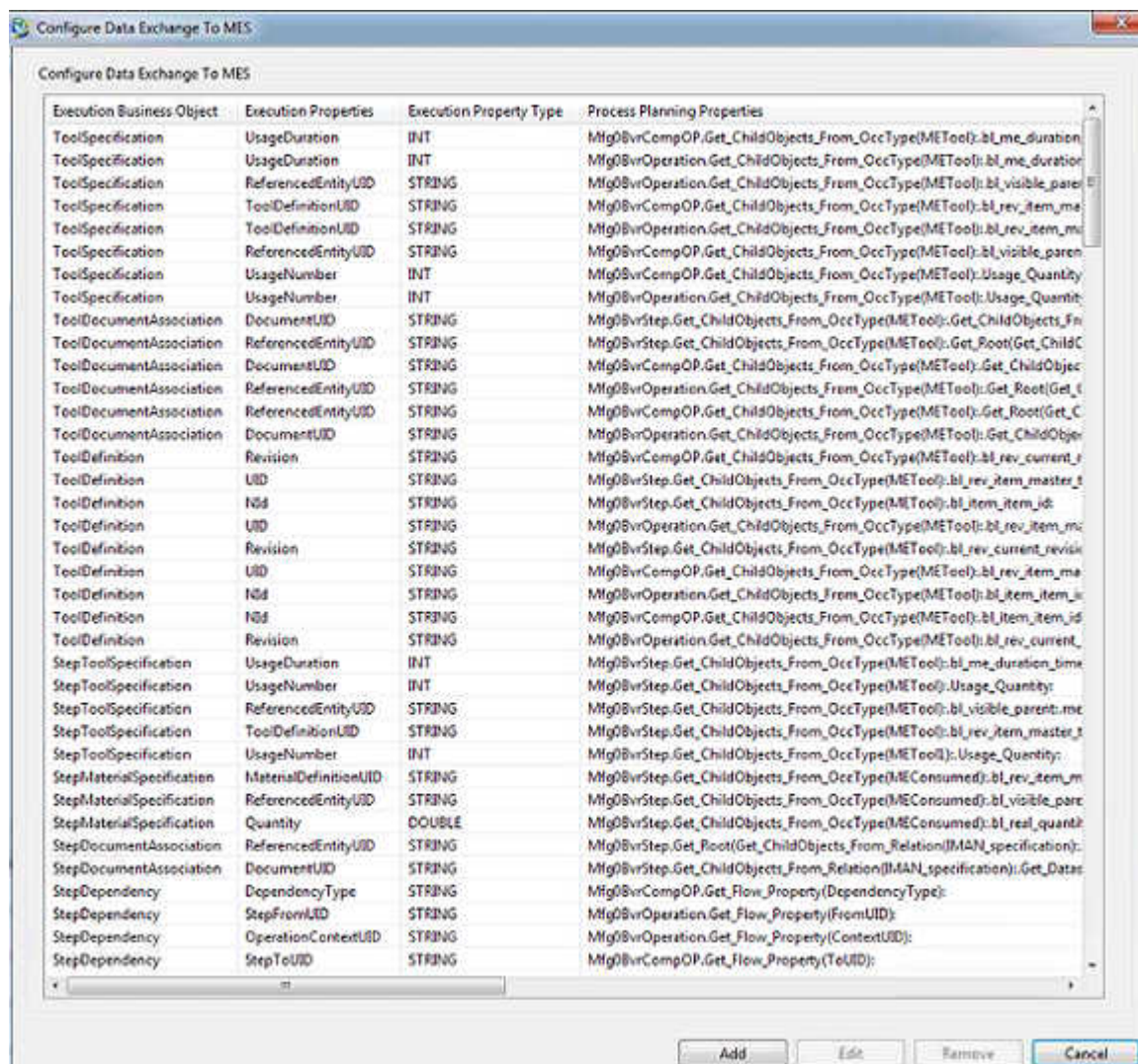
Define normalization mappings for CLM

In order to create an executable bill of process (BOP) for the CLM solution, you must use normalization mappings to map the execution properties in the SIMATIC IT data model to the process planning properties in the Teamcenter data model. A Teamcenter user with administrative privileges can use the **Configure Data Exchange to MES** dialog box to define and update the CLM normalization mappings.

1. In Manufacturing Process Planner, choose **Tools**→**MES**→**Configure Data Exchange to MES**.

This dialog box displays all the normalization mappings in a tabular format.

You can add a new mapping, edit an existing mapping, or select multiple mappings and remove them. The following sections describe how to add a mapping.



Defining execution properties

1. Click **Add** in the **Configure Data Exchange to MES** dialog box.
2. From the **Execution object** list in the **Configure Data Exchange** dialog box, select an appropriate option, for example, **AsPlannedBoP**.

The canonical model object names available in this list are defined in the **CLICanonicalModelObjectNames** preference.

3. From the **Execution property** list, select an appropriate option, for example, **PBOIdentID**.

You can also type a value. This value must match the value in the SIMATIC IT canonical data model.

4. From the **Execution property type** list, select an appropriate option, for example, **STRING**.

Defining process planning properties

In the **Process Planning Property** text box, you can add multiple clauses such as *attribute mapping* and *callback function*. While an attribute mapping clause consists of a business object and a subtype of the business object, the callback function clause consists of a business object, a registered callback to perform validation checks, and a function parameter such as an occurrence type.

Tip:

You can optionally add additional clauses to an existing clause without specifying a business object.

1. Specify the attribute mapping clause.
 - a. In the **Process Planning Property** text box, click **Add**.
 - b. From the **Select Clause** list, select **Attribute Mapping**.
 - c. From the **Business Object** list, select an appropriate option, for example, **BOMLine**.

The **CLi0NormalizationBusinessObjectTypes** preference specifies the list of Teamcenter business object types and their subtypes for defining normalization mappings. You can customize these values by adding a business object type to this preference or removing a business object type.

- d. From the **Property** list, select an appropriate option, for example, **bl_occurence_name**.
 - e. To add this attribute mapping to the **Process Planning Property** text box, click **OK**.
2. Specify the callback function clause.

- a. In the **Process Planning Property** text box, click **Add**.
- b. From the **Select Clause** list, select **Callback Function**.
- c. From the **Business Object** list, select an appropriate option, for example, **BOMLine**.
- d. From the **Callback Function** list, select an appropriate option, for example, **Get_ChildObjects_From_Occ_Type**.

The values in this list are available from **registered callbacks**.

- e. In the **Function Parameters** box, type the appropriate parameter, for example, **MEConsumed**. This is an occurrence type.
- f. To add this callback function to the **Process Planning Property** text box, click **OK**.

3. Specify the special string clause.

- a. In the **Process Planning Property** text box, click **Add**.
- b. From the **Select Clause** list, select **Special String**.
- c. To combine two process planning property values together using a special string, in the **Concatenation String** box, specify a separator such as the forward slash symbol (/).

Example:

```
Mfg0BvrProcess.bl_revision:.IMAN_MEWorkArea:.item_id:,"/",
bl_revision:.IMAN_MEWorkArea:.current_revision_id:
```

This results in **item_id/current_revision_id** such as **"000104/A"**.

4. To edit a property, click **Edit** in the **Process Planning Property** text box.
5. To delete the last line of a property, click **Remove** in the **Process Planning Property** text box.

4. Integrating with MES

Release an execution plan to the MES without an MCN

Perform a full release and later release selected BOP lines or changes to the MES without MCN

The data that is transferred to MES is contained in an execution plan. An execution plan consists of a process structure context with a root process and multiple processes containing small bills of processes (mini BOPs). When releasing an execution plan to the MES, you can perform a full release the first time and then release only delta changes. Teamcenter uses the release status to **track changes** and only updates based on their status, are sent to the MES.

After you perform a full release, you can release the selected BOP lines to the MES without applying a release status, or release only changes to the MES.

The following is the process to release an execution plan to the MES without manufacturing change notice (MCN):

1. **Create an execution plan containing the process structure and apply the release status MES Work In Progress to the process lines.**
2. **Apply effectivity and the status MES Pending Release to the execution plan.**
3. **(Optional) Assign product and manufacturing (PMI) to data in the execution plan.**
4. **Update the master order execution ID for the execution plan.**
5. **Validate MES data before releasing the execution plan to the MES.**
6. **Release the execution plan to the MES (classic integration)**
OR
Release the execution plan to the MES (CLM integration).
7. **(Optional after performing a full release) Release the selected BOP lines to the MES**
OR
(Optional after performing a full release) Release changes to the MES.

Create an execution plan

The object that is sent to the MES is an *execution plan*. An execution plan is a collaboration context of the type **Execution Plan**. The execution plan consists of a process, its compound operations, and execution steps as well as any consumed objects such as parts, resources, or work areas. Teamcenter automatically sends a notification when a changed master execution plan is released.

When creating an execution plan, you have several choices of how to build the structure.

- Top down:
 1. Create an empty collaboration context of type **Execution Plan**.
 2. Create an empty **Process** structure context within the execution plan.
 3. Fill the content of the new structure context by creating objects or copying and pasting existing objects.
- Bottom up:
 1. Open a process whose structure conforms with the requirements (for example, structure, types, level, assignment).
 2. Save the process as an **Process** structure context.
 3. Save all structure contexts in the session to a new execution plan collaboration context or move structure contexts into an existing execution plan.

To create the objects in an execution plan:

Object	Is a	Create by choosing
Execution plan	Collaboration context	File → New → Collaboration Context → Execution plan
Process root	Process	File → New → Process (Process) or click  in the process structure view.
Operation	Operation	File → New → Operation or click  in the process structure view.
Compound operation	Process	Click  in the process structure view.
Execution step	Operation	Click  in the process structure view.

To perform a validation check on an execution plan:

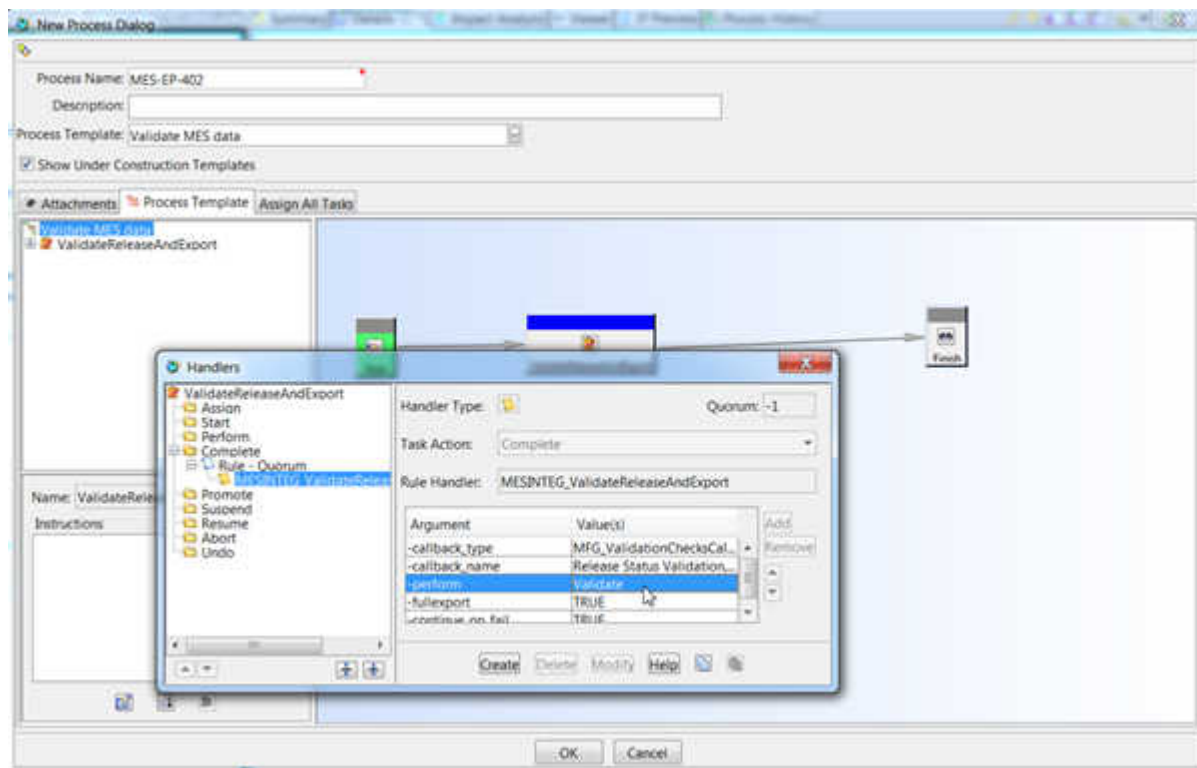
1. Create a workflow template for the **MESINTEG_ValidateReleaseAndExport** rule handler with **Validate** as the value for the **-perform** argument.

The **MESINTEG_ValidateReleaseAndExport** rule handler performs validation checks for MES Integration.

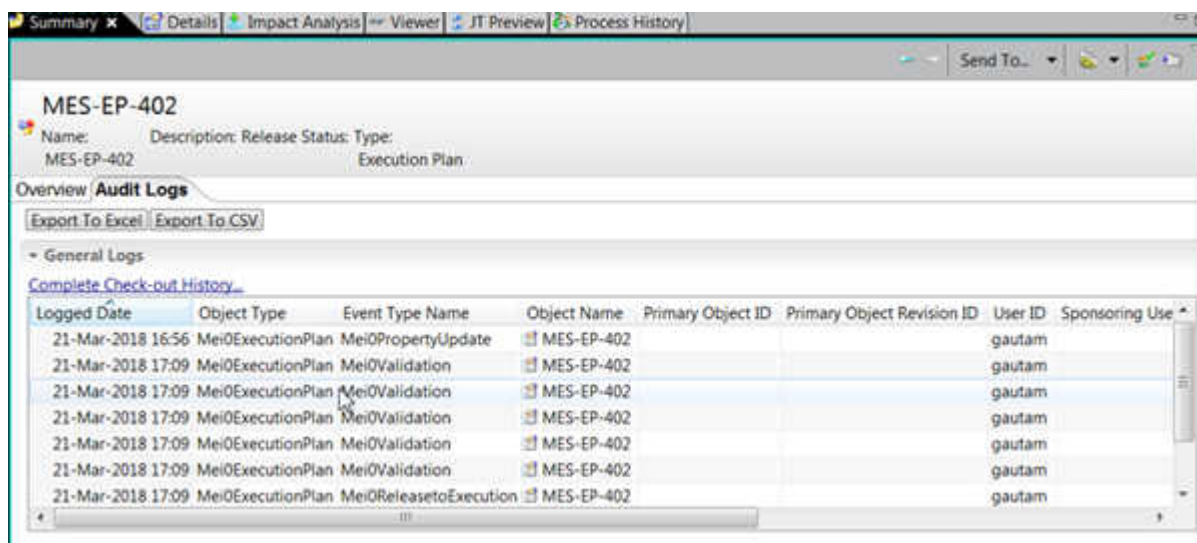
Note:

Release to MES using the **MESINTEG_ValidateReleaseAndExport** rule handler is *not* supported for plant BOPs.

2. In My Teamcenter, select the execution plan and click **File** → **New** → **Workflow Process**.
3. In **New Process Dialog**, specify a process name and select the workflow process template you created in step 1 and click **OK**.



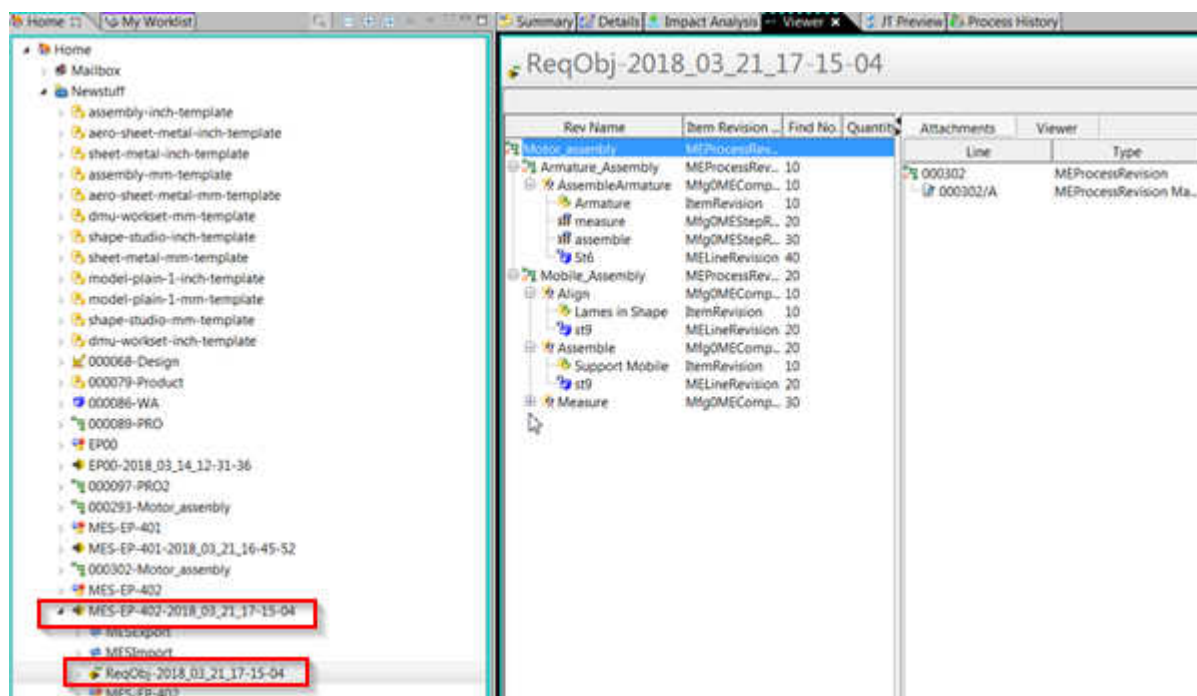
4. To view the validation logs, select the execution plan from the **Home** folder in My Teamcenter, and in the **Summary** tab, click **Audit Logs**.



5. If there are no validation errors, you can use another workflow process to release the execution plan to MES Integration.

After the workflow process is complete, you can refresh the folder in the **Home** view of My Teamcenter. You can expand the execution plan to view the application interface objects and request objects created by the process after performing a release.

The **ReqObj** object contains the structure sent to MES Integration.



6. To view the logs, select the **Summary** tab and click **Audit Logs**.

Assign product and manufacturing information to data in an execution plan

Product and manufacturing information (PMI) is used in 3D CAD or collaborative product development systems to transfer information about the design of a product's components for manufacturing. Specifically, PMI contains geometric dimensions and tolerances, 3D annotation (text), surface finish, and material specifications.

You can use the **Data Collection** menu in Manufacturing Process Planner to open the web page of the third-party integration. Optionally, you can specify PMI attributes relevant to your data in the execution plan. The web page is used to manage ALT forms associations that contain the PMI attributes relevant to the shop floor operator. The third-party integrator manages this web page and is responsible for creating and maintaining ALT forms associations.

Note:

The procedure for assigning PMI data in this topic is specific to the integration partner, SIMATIC IT. Other integration partners may implement this differently.

At your site, the system administrator can specify the URL for the web page using the **MEMESManageSITDCD** preference and show or hide the **Data Collection** menu using the **MEMESEnableDataCollection** preference.

1. Select a compound operation, execution step, or an OP type in an execution plan.
2. To open the third-party integration web page, select **Tools→MES→Data Collection**.
3. (Optional) Assign or remove ALT forms using the web page.

The Access Manager (AM) rule tree determines the privileges on objects in the database. You can assign ALT forms to objects only if you have write privileges, and this depends on the access control list (ACL) rules.

4. (Optional) Specify the PMI attributes relevant to your data in the execution plan using the ALT forms in the web page.

At your site, the system administrator can modify the default PMI properties, if required, using the **MEMESPMIProperties** preference. This preference contains runtime properties defined on the BOMline.

By default, the following properties are available: **Characteristics Type**, **mci0PMIDescription**, **PMI Dimension Type**, **PMI Inspection Type**, **PMI Lower Spec Limit**, **PMI Upper Spec Limit**, **PMI NominalValue**, **PMI GDT Text**, **PMI GDT Code**, and **PMI Frequency**.

You can add or remove the default preference values depending on the customization done at your site.

Apply effectivity and release status to an execution plan without MCN

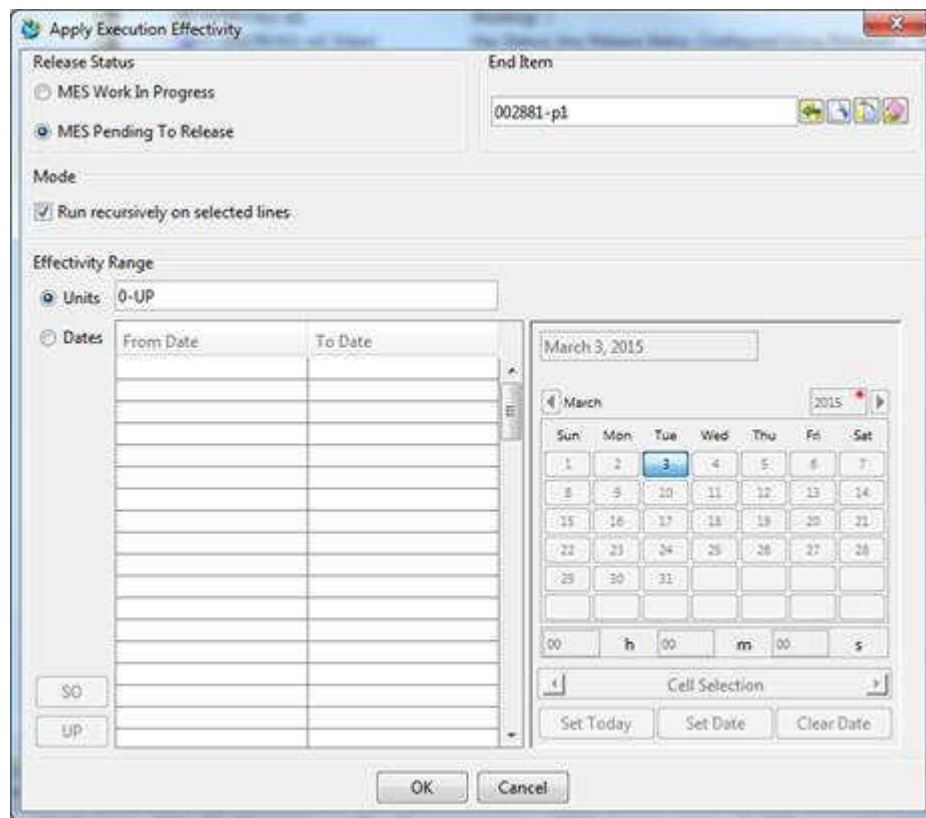
You designate which processes and operations in a master are included in particular order by specifying the effectivity of the individual lines in the process structure.

1. Select a compound operation or execution step and click .

Teamcenter displays the **Apply Execution Effectivity** dialog box.

2. Type the unit or unit range for which this line is effective. This represents the order number in which this line should be included.

You can type a split effectivity value such as **5-10**, **3-UP**, or **5-SO**.



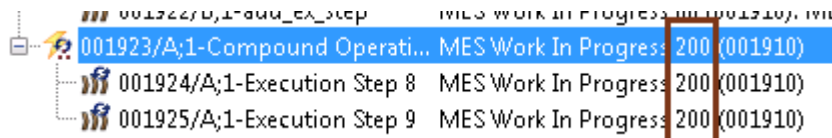
Note:

Do not use the **Dates** option in the **Effectivity Range** area for MES Integration.

3. Select **Run recursively on selected objects** to apply the same effectivity to all the child lines.
4. If this line (and its children, if you choose to run this command recursively) are ready for release, select **Mark for Release**.

- Click **OK**.

Teamcenter applies the order effectivity on the selected line and its children (if you chose that option).



Note:

In this example, if you choose **Mark for Release** in the **Apply Order Effectivity** dialog box, the status is switched from **MES Work In Progress** to **MES Pending To Release**.

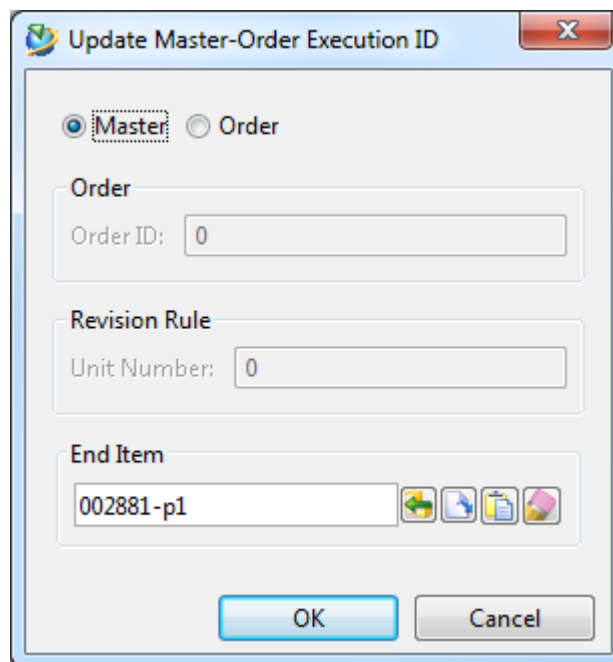
Update the master order for an execution plan

You must set the order effectivity for an execution plan before you release the execution plan to MES.

Note:

The following procedures are not relevant if you want to release a plant structure to production.

- In the **Collaboration Context Tree** view, select the execution plan and click .



- For the order, type an ID in the **Order ID** box.

You can specify the same number as that of the order ID if the order ID contains only numeric characters.

The system saves the order ID as a property on the execution plan collaboration context.

3. For the revision rule, type a unit effectivity in the **Unit Effectivity** box.

You can specify unit effectivity as an integer or a range that includes the UP or SO string, such as **5-10**, **3-UP**, or **5-SO**.

4. (Optional) In the **End Item** box, do one of the following:

- Type an end item.
- Click  and choose a most recently used end item.
- Click , type a name, and choose **Find** to find an end item.

In the **Set Order ID** dialog box, the system displays the default end item. This is the same as the end item in the revision rule. If it does not exist, the system displays the top line item as the end item by default.

If you change the end item, the system saves it as part of the revision rule in the configuration context.

Validate MES data

You must validate whether a particular structure contains the necessary data in the proper format before you transfer the structure to the manufacturing execution system or production systems.

1. Select a process structure or one or more process or operation lines.
2. Choose **Tools**→**MES**→**Validate for MES**.
3. In the **Validate for MES** dialog box, select the structure you want to validate. You can add structures by selecting them in the structure view and clicking . You can remove structures from the list by clicking .

You can only add structures from the same process structure context.

4. Select the type of validity check you want to perform.

You can use the check box to select all the validity checks or remove all the selections with one click.

Note:

The validations displayed in the **Validate for MES** dialog box are registered using the **MEMESValidationList** preference.

5. Click **OK**.

If Teamcenter cannot validate the data, the **MES Validation Results** view is displayed listing the issues.

If Teamcenter validates the data and no problems are found, but the **MES Validation Results** view is open due to a previous validation, the view displays a success message.

6. In the **MES Validation Results** view, select each line to see what problems exist with the data and resolve each problem. The **Validation Results** list shows which validity check produced the error listed in the **Description** column.

As you solve the problems, remove the line from the list by clicking .

Removing lines can help make the list of problems more manageable. If you erroneously remove a line, there is no way to retrieve it. You must rerun the validation.

7. (Optional) Export the MES data validation results to a Microsoft Excel file by clicking  in the **MES Validation Result View**.

Running a data validation can be time-consuming, depending on how much data you select. Rerunning the validation may not be practicable. If you interrupt your work, you can save the results to an Excel file and work through each line until you solve all the problems before you rerun the validation. In addition, you can use the Excel file to filter data according to issue severity.

Teamcenter stores the data in a **.CSV** file that you can open in Excel.

Release an execution plan to the MES (classic integration)

1. Select a valid execution plan from the **Collaboration Context Tree** view in Manufacturing Process Planner or Part Planner.
2. Choose the **Tools**→**MES**→**Release to Execution** menu command or choose **Release to Execution** from the shortcut menu.
3. Choose the workflow that you want to use to transfer the data.

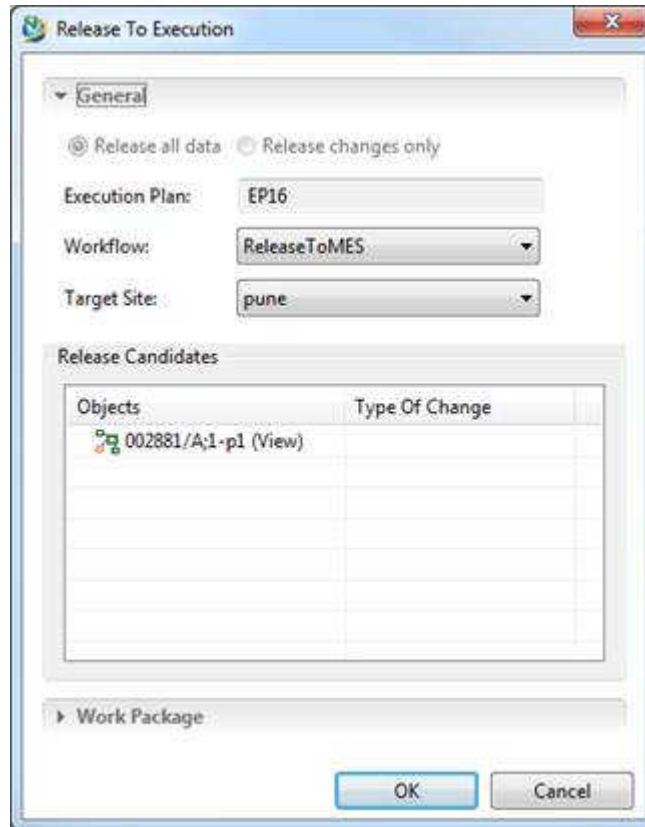
The list of available workflows is stored in the **MEMESWorkflowAndApplicationInterfaceType** preference. The **ReleaseToMES** workflow is delivered with the product.

4. Click the arrow beside the **Target Sites** box to select the sites to which you want to send the data.

The list of available sites is stored in the **MEMESSites** preference.

The line you see in the **Objects** list is the top line for a full release and delta lines are displayed for a delta release.

Teamcenter displays the name of the application interface object as the name of the work package.



5. Click **OK**.

Even if you already validated the selected data, Teamcenter validates the data again as the data may have changed since the validation. It then initiates the workflow and creates the work package. The **Release to Execution** command creates the application interface and the workflow adds an application interface request to the application interface. You can view the status of the AI requests in the application interface viewer.

6. (Optional) View the log file to see any errors generated by the workflow. The log file is saved as a dataset of type **TEXT** and added to the workflow as a reference attachment in the task attachments **References** folder. To view this log file:
 - a. Select the execution plan and choose **Send To→ Workflow Viewer**. **Workflow Viewer** displays the completed workflow process.
 - b. Right-click the validation task and choose **Properties**.

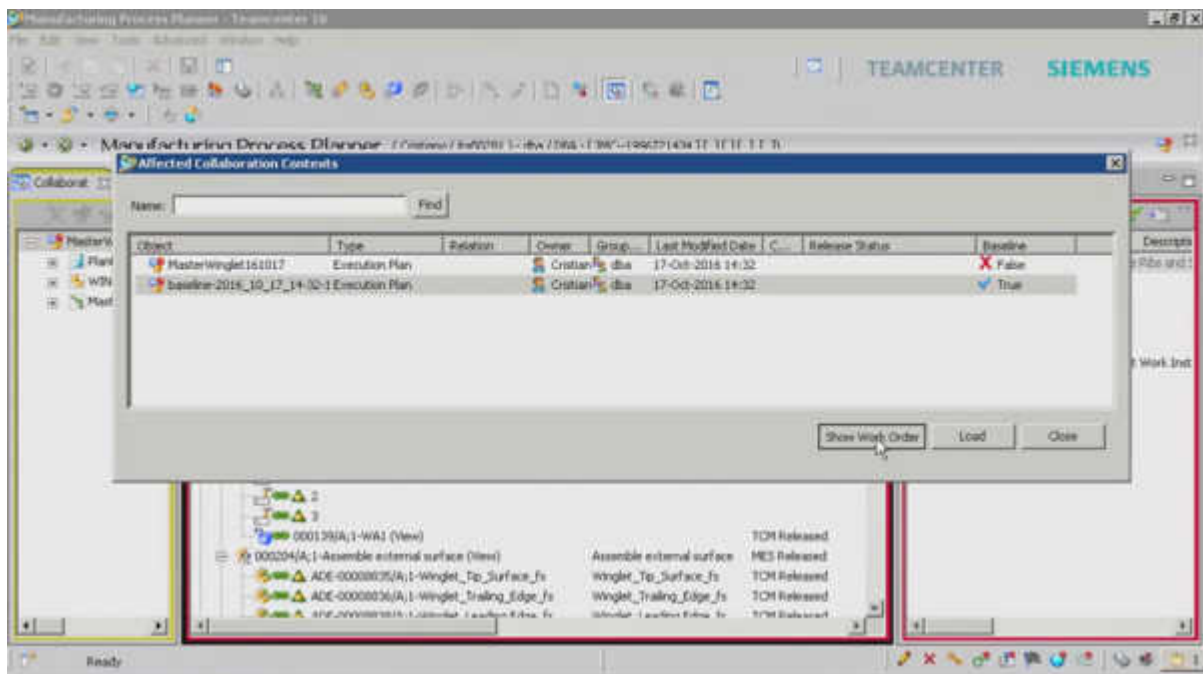
- c. In the **References** box, double-click the log file to open it.

Release an execution plan to the MES (CLM integration)

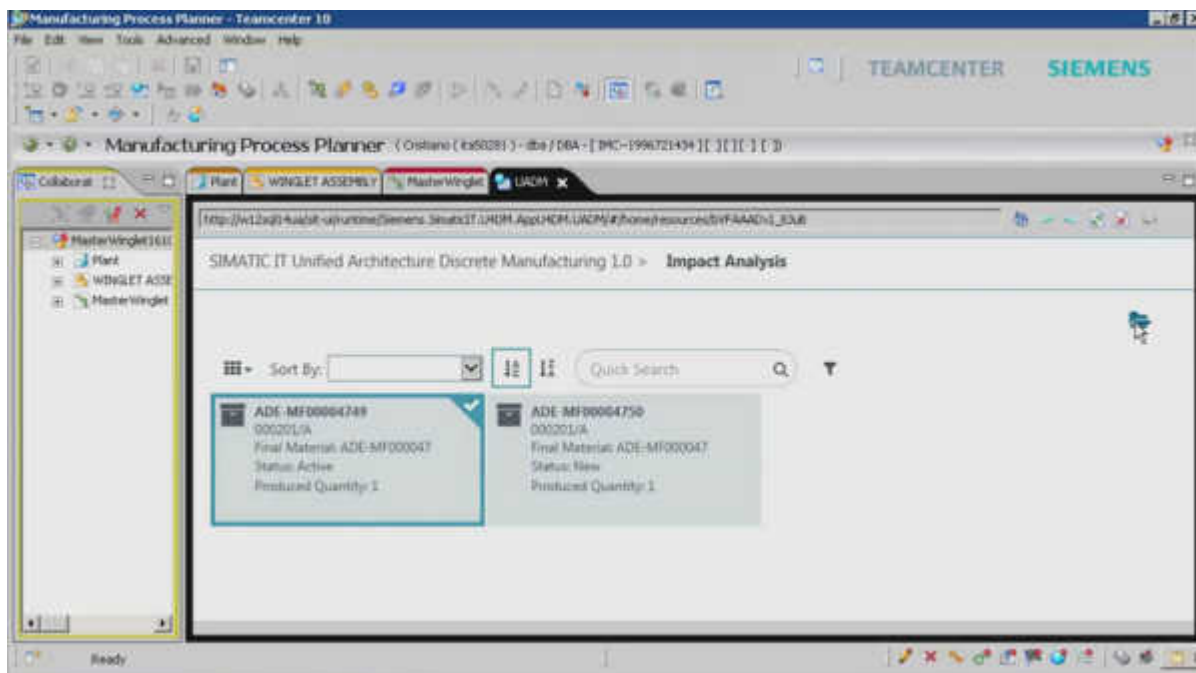
View related work orders to assess the impact of change on the shop floor work order in progress

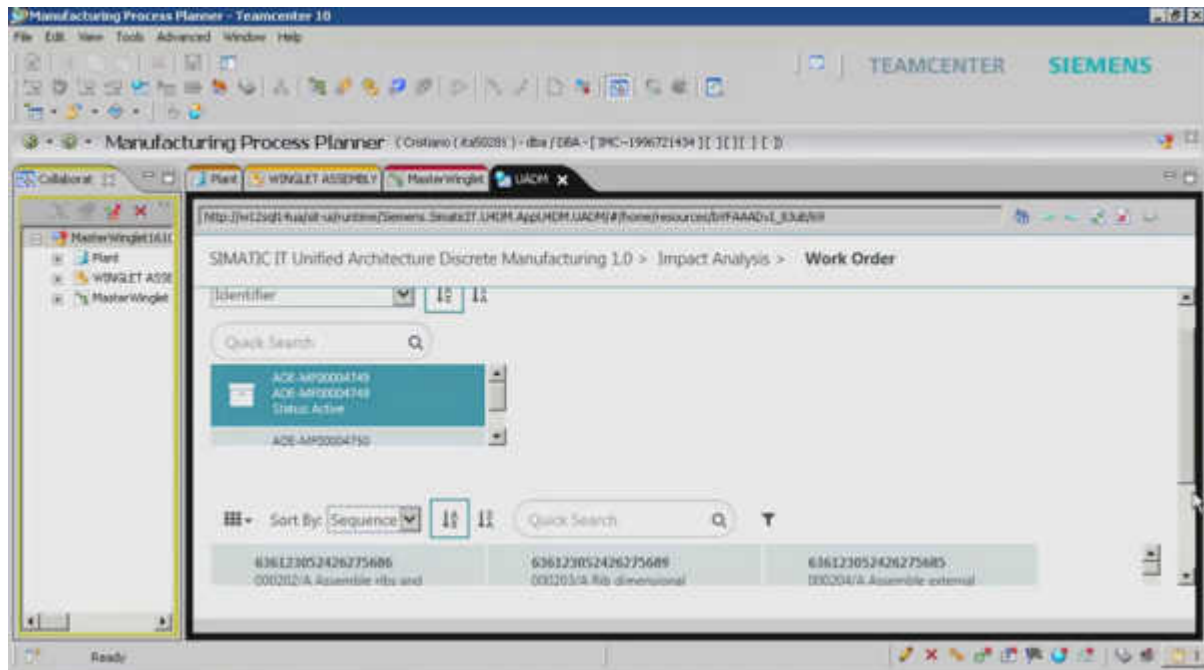
Before you release the changes to the MES Integration, you can view related work orders to assess the impact of change on the shop floor work order in progress.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. Right-click a line in a structure and choose **Tools**→**Find Affected Collaboration Contexts**.



3. Select the appropriate baseline in the **Affected Collaboration Contexts** dialog box and click **Show Work Order**.





Release an execution plan (CLM integration)

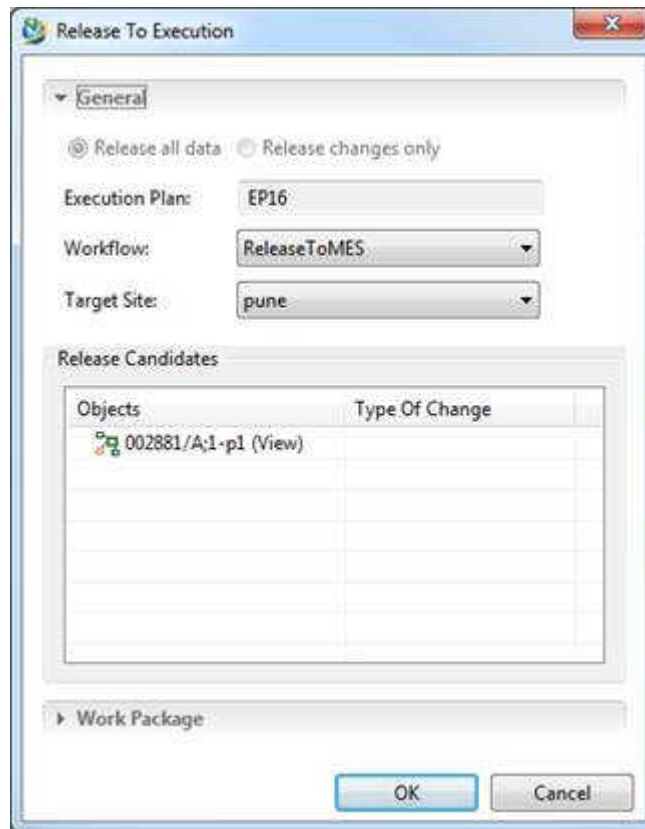
1. Select a valid execution plan from the **Collaboration Context Tree** view in Manufacturing Process Planner or Part Planner.
2. Choose the **Tools→MES→Release to Execution** menu command or choose **Release to Execution** from the shortcut menu.
3. Choose the workflow that you want to use to transfer the data.

The list of available workflows is stored in the **MEMESWorkflowAndApplicationInterfaceType** preference. The **ReleaseToMES** workflow is delivered with the product.

4. Click the arrow beside the **Target Sites** box to select the sites to which you want to send the data.

The list of available sites is stored in the **MEMESSites** preference.

The line you see in the **Objects** list is the top line for a full release and delta lines are displayed for a delta release.



5. Click **OK**.

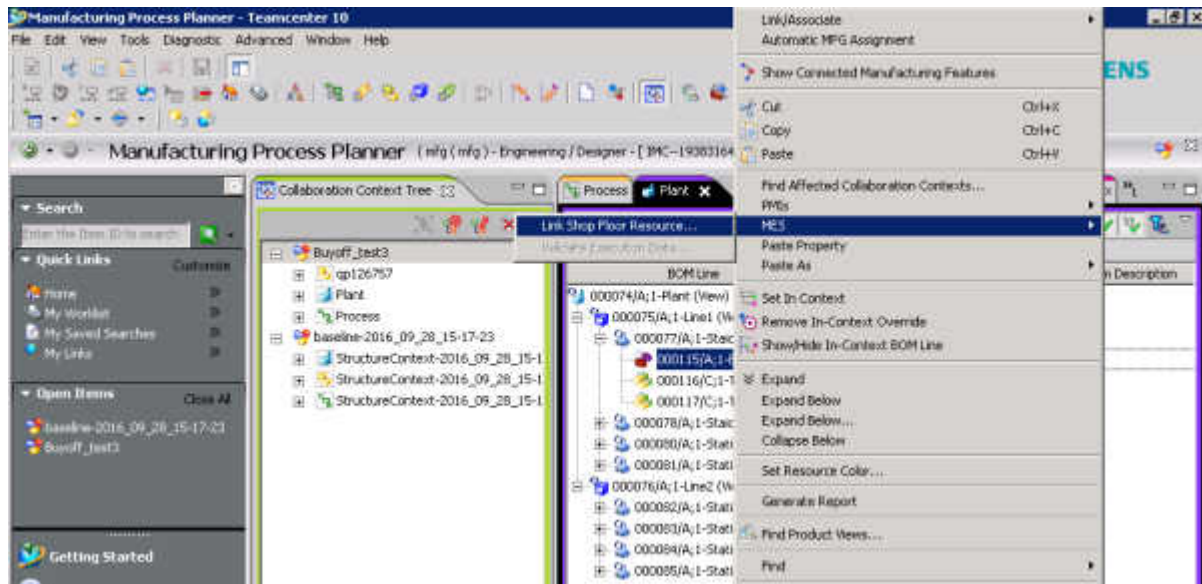
Even if you already validated the selected data, Teamcenter validates the data again as the data may have changed since the validation. It then initiates the workflow and creates a baseline and record object. The **Release to Execution** command exports the BOP in a lightweight JSON format and persists this data as an As Planned BOP in SIMATIC IT.

6. (Optional) **View the log file available on the record object.**

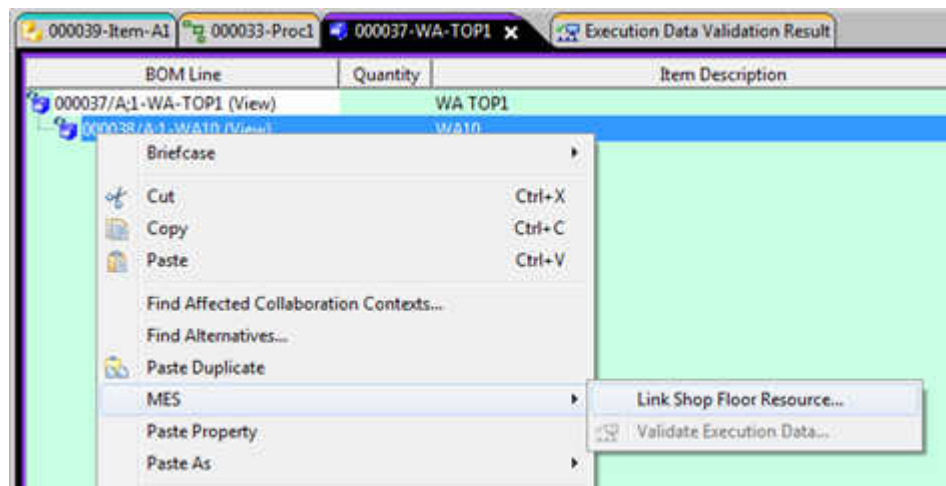
Link the plant structure in Teamcenter to the SIMATIC IT plant structure

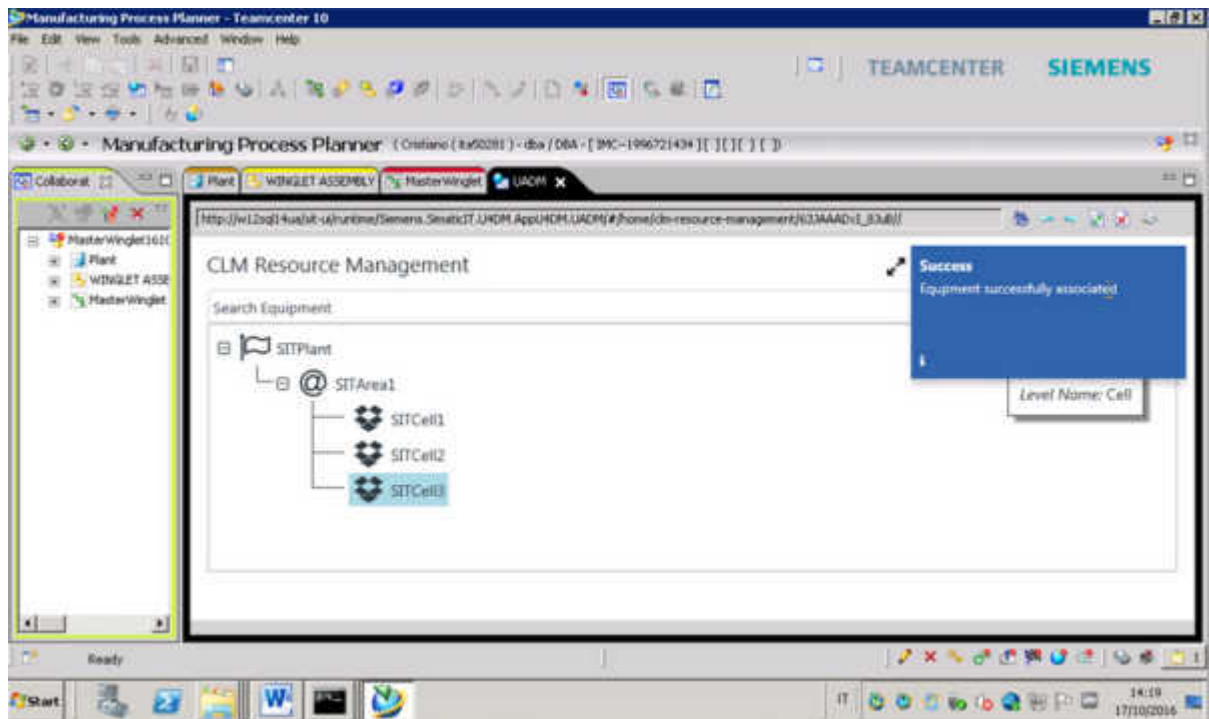
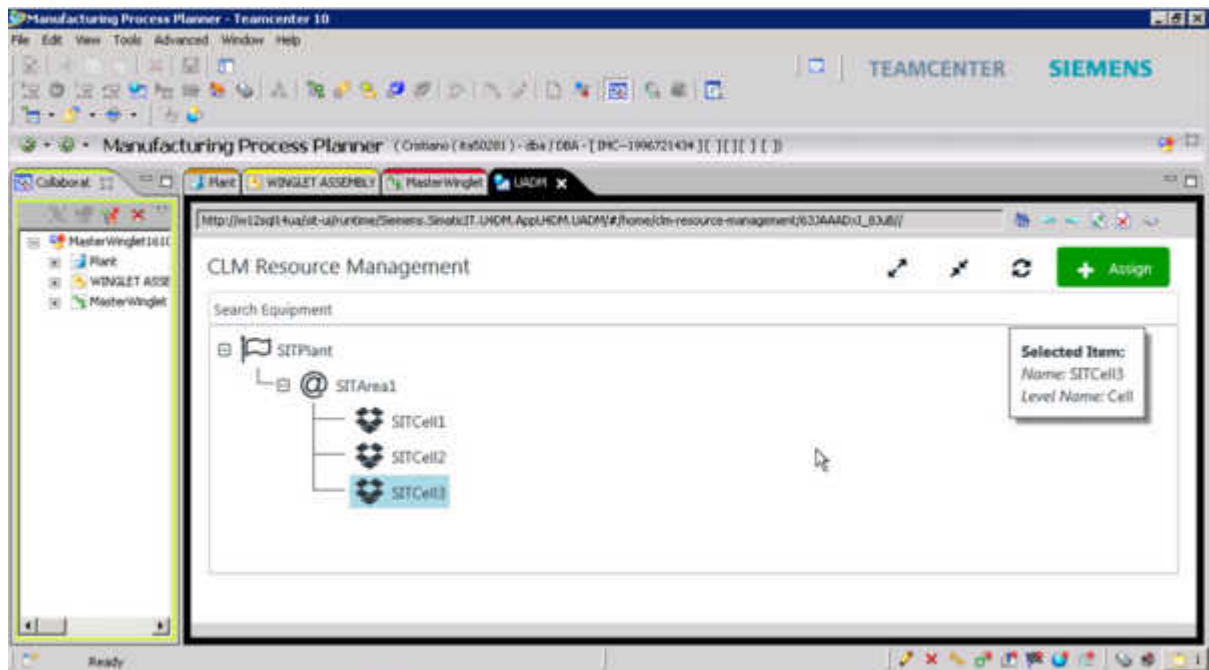
You can link Teamcenter objects such as work area and resources to SIMATIC IT plant model.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. Click the Plant view and select a work area or a resource line (for example, equipment).
3. Right click and choose **MES→Link Shop Floor Resource** to open the SIMATIC IT web page.



4. Select the work area or resource line to which you want to link to in the SIMATIC IT plant structure and click **Assign**.





Release selected BOP lines to the MES

You can send selected BOP lines from the structure to the MES Integration. The system then does not apply any release status to the selected lines, unlike when you release an execution plan to the MES. However, you must perform a full release once before sending the selected BOP lines to the MES.

The system administrator must enable the **MEMESSendWorkflows** site preference and set it to the **SendToMES** default value before you can release selected BOP lines to the MES. In addition, the workflow administrator must set the **-export_as_fai=true** argument to the **MESINTEG_ValidateReleaseAndExport** rule handler. If this property is set to **true**, the work package is considered as a part of the **Send to MES** command.

1. Select a valid execution plan from the **Collaboration Context Tree** view in Manufacturing Process Planner or Part Planner.
2. Select the BOP lines in the structure you want to send to the MES.
3. Choose the **Tools→MES→Send Line(s) to MES** menu command or choose **Send Line(s) to MES** from the shortcut menu.

Note:

You must perform a full release once before sending the selected BOP lines to the MES.

4. Choose the **SendToMES** workflow to transfer the data.

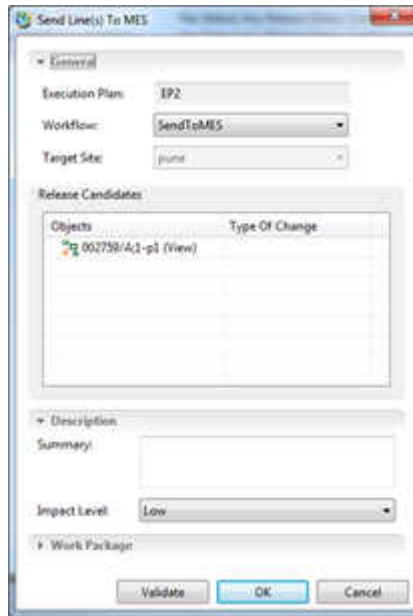
Tip:

The list of available workflows is stored in the **MEMESSendWorkflows** preference.

5. Click the arrow near the **Target Sites** box to select the sites to which you want to send the data.

The list of available sites is stored in the **MEMESSites** preference.

Teamcenter displays the name of the application interface object as the name of the work package.



6. Click **OK**.

Even if you have already validated the selected data, Teamcenter validates the data again as the data may have changed since the validation. It then initiates the workflow and creates the work package. The **Release to Execution** command creates the application interface and the workflow adds an application interface request to the application interface. You can view the status of the AI requests in the application interface viewer.

7. (Optional) View the log file to view any errors generated by the workflow. The log file is saved as a dataset of type **TEXT** and added to the workflow as a reference attachment in the task attachments **References** folder. To view this log file:
 - a. Select the execution plan and choose **Send To → Workflow Viewer**. **Workflow Viewer** displays the completed workflow process.
 - b. Right-click the validation task and choose **Properties**.
 - c. In the **References** box, double-click the log file to open it.

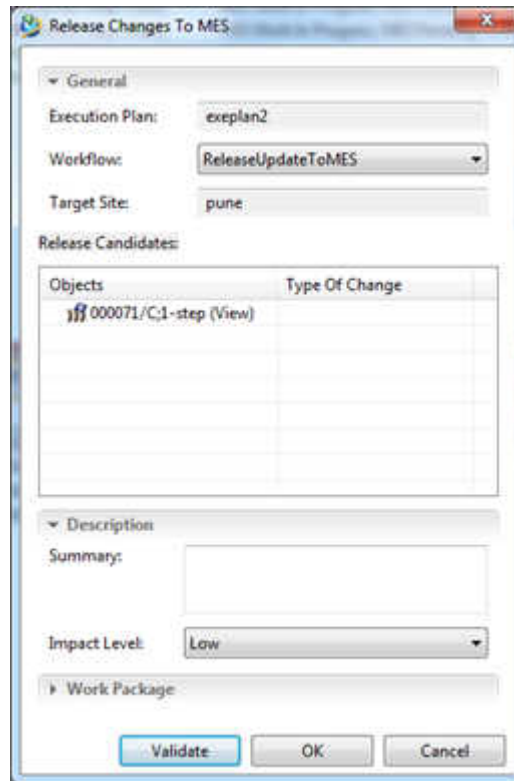
Release changes to the MES

1. Select an execution plan and click .

Teamcenter checks whether the execution plan has already been released to the MES and displays the **Release Options** dialog box.

2. Select **Release changes only** and click **OK**.

Teamcenter displays the **Release changes to MES** dialog box with the name of the execution plan and the name of a workflow.



To track changes for each update to the MES Integration system, the **Release changes to MES** dialog box displays all modifications done to a consumption or work area under an incremental change.

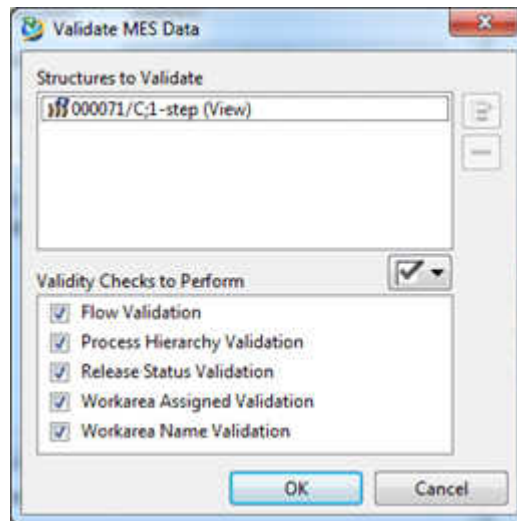
The workflows displayed in this list are those found in the **MEMESUpdateWorkflows** preference in the order that they appear in that preference. In addition, the **Release Candidates** list displays all the lines found in the selected structure that have an **MES Pending To Release** status. The **Type of Change** column displays the nature of the change. If it is the result of an incremental change, you can click the entry in this column to see the change. If the change is the result of revising the line, the cell remains empty.

3. (Optional) Type a description of the work package and choose an impact level.
4. (Optional) Open the **Work Package** section to display the name and creation date of the work package.

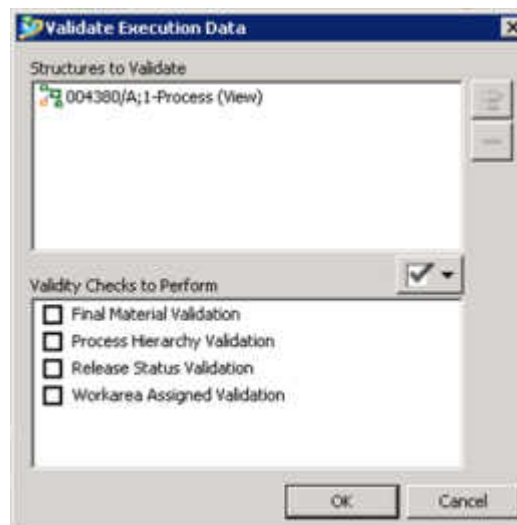
You can use the name to search for the work package in My Teamcenter.

5. To validate only the incremental changes (delta), click the **Validate** button to open the **Validate MES Data** dialog box.

- Using classic integration:



- Using CLM integration:



Release an execution plan to the MES using an MCN

Perform a full release and then delta changes to the MES using an MCN

The data that is transferred to the MES is contained in an execution plan. An execution plan consists of a process structure context with a root process and multiple processes containing small bills of processes (mini BOPs). When releasing an execution plan to the MES, you can perform a full release the first time and then release only delta changes. Teamcenter uses the release status to **track changes** and only updates based on their status, are sent to the MES.

After you perform a full release, you can release the selected BOP lines to the MES without applying a release status, or release only changes to the MES.

The following is the process to release an execution plan to the MES using an MCN:

1. **Create an execution plan containing the process structure.**
2. **Set the MEMCN_management_enabled preference value to True.**
3. Create an MCN.
4. **On the MCN, set the release status to MES Work In Progress and apply date or unit effectivity.**
5. **(Optional) Assign product and manufacturing (PMI) to data in an execution plan.**
6. **Apply effectivity and release status to the process structure.**

Note:

Perform this step only if the **MEMCN_release_revision_mode** preference value is set to any other value other than **Share**.

7. **On the MCN, set the release status to MES Pending to Release.**
8. **Validate MES data before releasing an execution plan to the MES.**
9. **Release the execution plan to the MES (classic integration)**
OR
Release the execution plan to the MES (CLM integration).

If you make minor changes after performing a full release, create a new MCN, and then release the changes to the MES.

Create an execution plan

The object that is sent to the MES is an *execution plan*. An execution plan is a collaboration context of the type **Execution Plan**. The execution plan consists of a process, its compound operations, and execution steps as well as any consumed objects such as parts, resources, or work areas. Teamcenter automatically sends a notification when a changed master execution plan is released.

When creating an execution plan, you have several choices of how to build the structure.

- Top down:
 1. Create an empty collaboration context of type **Execution Plan**.

2. Create an empty **Process** structure context within the execution plan.
 3. Fill the content of the new structure context by creating objects or copying and pasting existing objects.
- Bottom up:
 1. Open a process whose structure conforms with the requirements (for example, structure, types, level, assignment).
 2. Save the process as an **Process** structure context.
 3. Save all structure contexts in the session to a new execution plan collaboration context or move structure contexts into an existing execution plan.

To create the objects in an execution plan:

Object	Is a	Create by choosing
Execution plan	Collaboration context	File → New → Collaboration Context → Execution plan
Process root	Process	File → New → Process (Process) or click  in the process structure view.
Operation	Operation	File → New → Operation or click  in the process structure view.
Compound operation	Process	Click  in the process structure view.
Execution step	Operation	Click  in the process structure view.

To perform a validation check on an execution plan:

1. Create a workflow template for the **MESINTEG_ValidateReleaseAndExport** rule handler with **Validate** as the value for the **-perform** argument.

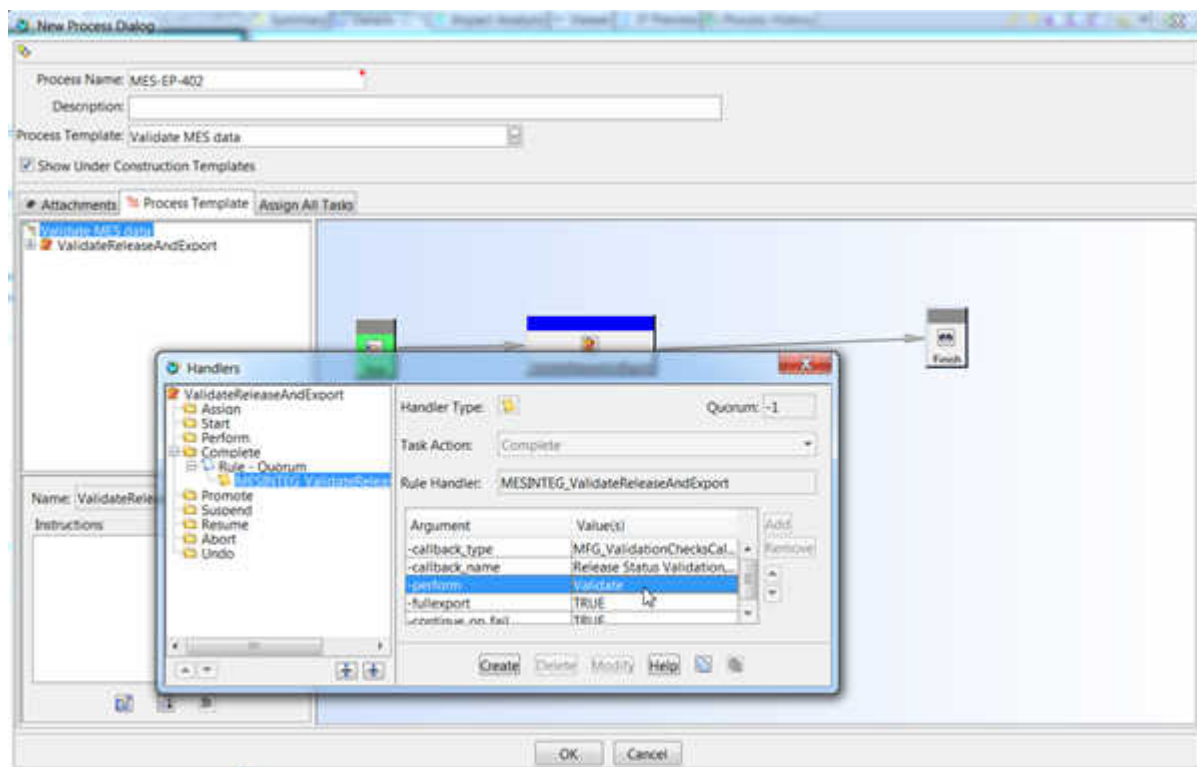
The **MESINTEG_ValidateReleaseAndExport** rule handler performs validation checks for MES Integration.

Note:

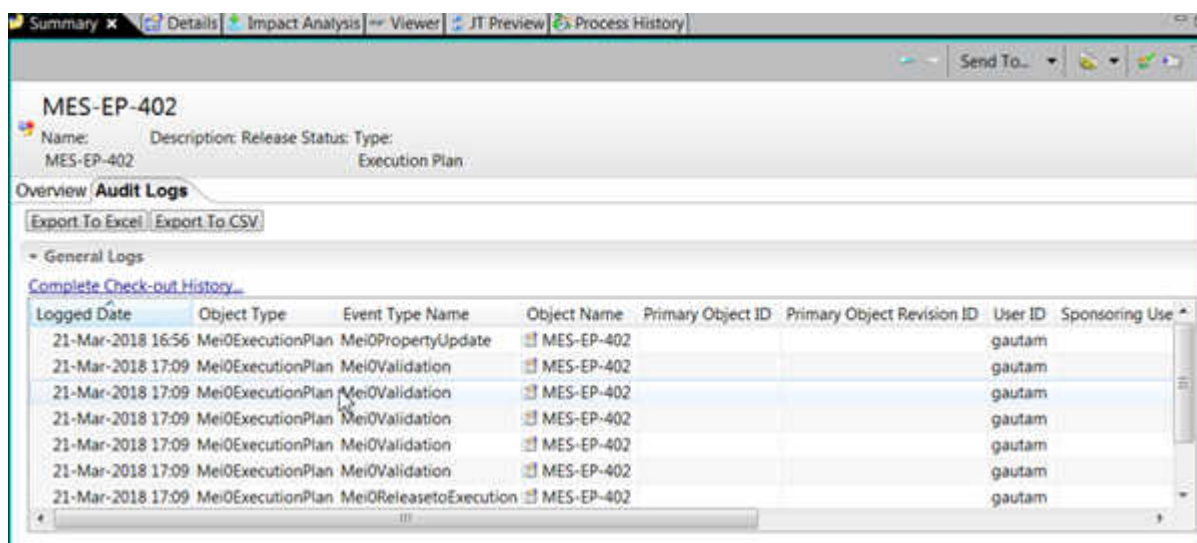
Release to MES using the **MESINTEG_ValidateReleaseAndExport** rule handler is *not* supported for plant BOPs.

2. In My Teamcenter, select the execution plan and click **File** → **New** → **Workflow Process**.

3. In **New Process Dialog**, specify a process name and select the workflow process template you created in step 1 and click **OK**.



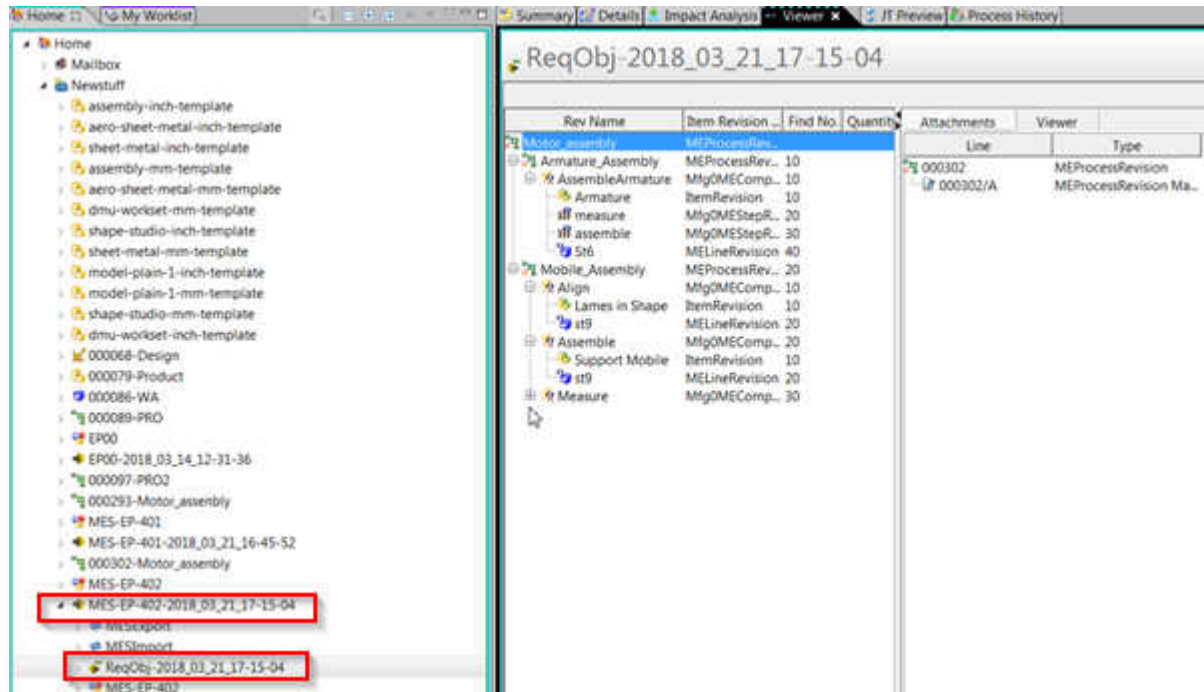
4. To view the validation logs, select the execution plan from the **Home** folder in My Teamcenter, and in the **Summary** tab, click **Audit Logs**.



5. If there are no validation errors, you can use another workflow process to release the execution plan to MES Integration.

After the workflow process is complete, you can refresh the folder in the **Home** view of My Teamcenter. You can expand the execution plan to view the application interface objects and request objects created by the process after performing a release.

The **ReqObj** object contains the structure sent to MES Integration.



6. To view the logs, select the **Summary** tab and click **Audit Logs**.

Setting the MCN preference values

MCNs are engineering change notices that are specifically intended for manufacturing rather than engineering needs. The implementation engineer enables the change notice revision (CNR) related functionality in Manufacturing Process Planner by setting the **MEMCN_management_enabled** preference to **True**.

In this case, you must release an execution plan with an MCN. If the preference is set to **False**, you release the execution plan without the MCN.

Apply the effectivity range and the release status in the **Apply Execution Effectivity** dialog box. The **Apply Execution Effectivity** command is enabled or disabled based on the **MEMCN_management_enabled** and the **MEMCN_release_revision_mode** preference values.

- The **Apply Execution Effectivity** dialog box displays only unit effectivity values if the **MEMCN_management_enabled** preference value is set to **False**.
- The **Apply Execution Effectivity** dialog box is populated with the release status and the date or unit effectivity only if the change object is set and has the release status and the date or unit effectivity.

- The **Apply Execution Effectivity** dialog box is populated with just the default release status, and without values for the date or unit effectivity if no change object is set.

Assign product and manufacturing information to data in an execution plan

Product and manufacturing information (PMI) is used in 3D CAD or collaborative product development systems to transfer information about the design of a product's components for manufacturing. Specifically, PMI contains geometric dimensions and tolerances, 3D annotation (text), surface finish, and material specifications.

You can use the **Data Collection** menu in Manufacturing Process Planner to open the web page of the third-party integration. Optionally, you can specify PMI attributes relevant to your data in the execution plan. The web page is used to manage ALT forms associations that contain the PMI attributes relevant to the shop floor operator. The third-party integrator manages this web page and is responsible for creating and maintaining ALT forms associations.

Note:

The procedure for assigning PMI data in this topic is specific to the integration partner, SIMATIC IT. Other integration partners may implement this differently.

At your site, the system administrator can specify the URL for the web page using the **MEMESManageSITDCD** preference and show or hide the **Data Collection** menu using the **MEMESEnableDataCollection** preference.

1. Select a compound operation, execution step, or an OP type in an execution plan.
2. To open the third-party integration web page, select **Tools**→**MES**→**Data Collection**.
3. (Optional) Assign or remove ALT forms using the web page.

The Access Manager (AM) rule tree determines the privileges on objects in the database. You can assign ALT forms to objects only if you have write privileges, and this depends on the access control list (ACL) rules.

4. (Optional) Specify the PMI attributes relevant to your data in the execution plan using the ALT forms in the web page.

At your site, the system administrator can modify the default PMI properties, if required, using the **MEMESPMIProperties** preference. This preference contains runtime properties defined on the BOMline.

By default, the following properties are available: **Characteristics Type**, **mci0PMIDescription**, **PMI Dimension Type**, **PMI Inspection Type**, **PMI Lower Spec Limit**, **PMI Upper Spec Limit**, **PMI NominalValue**, **PMI GDT Text**, **PMI GDT Code**, and **PMI Frequency**.

You can add or remove the default preference values depending on the customization done at your site.

Validate MES data

You must validate whether a particular structure contains the necessary data in the proper format before you transfer the structure to the manufacturing execution system or production systems.

1. Select a process structure or one or more process or operation lines.
2. Choose **Tools**→**MES**→**Validate for MES**.
3. In the **Validate for MES** dialog box, select the structure you want to validate. You can add structures by selecting them in the structure view and clicking . You can remove structures from the list by clicking .

You can only add structures from the same process structure context.

4. Select the type of validity check you want to perform.

You can use the check box to select all the validity checks or remove all the selections with one click.

Note:

The validations displayed in the **Validate for MES** dialog box are registered using the **MEMESValidationList** preference.

5. Click **OK**.

If Teamcenter cannot validate the data, the **MES Validation Results** view is displayed listing the issues.

If Teamcenter validates the data and no problems are found, but the **MES Validation Results** view is open due to a previous validation, the view displays a success message.

6. In the **MES Validation Results** view, select each line to see what problems exist with the data and resolve each problem. The **Validation Results** list shows which validity check produced the error listed in the **Description** column.

As you solve the problems, remove the line from the list by clicking .

Removing lines can help make the list of problems more manageable. If you erroneously remove a line, there is no way to retrieve it. You must rerun the validation.

7. (Optional) Export the MES data validation results to a Microsoft Excel file by clicking  in the **MES Validation Result View**.

Running a data validation can be time-consuming, depending on how much data you select. Rerunning the validation may not be practicable. If you interrupt your work, you can save the results to an Excel file and work through each line until you solve all the problems before you rerun the validation. In addition, you can use the Excel file to filter data according to issue severity.

Teamcenter stores the data in a **.CSV** file that you can open in Excel.

Release an execution plan to the MES (classic integration)

1. Select a valid execution plan from the **Collaboration Context Tree** view in Manufacturing Process Planner or Part Planner.
2. Choose the **Tools**→**MES**→**Release to Execution** menu command or choose **Release to Execution** from the shortcut menu.
3. Choose the workflow that you want to use to transfer the data.

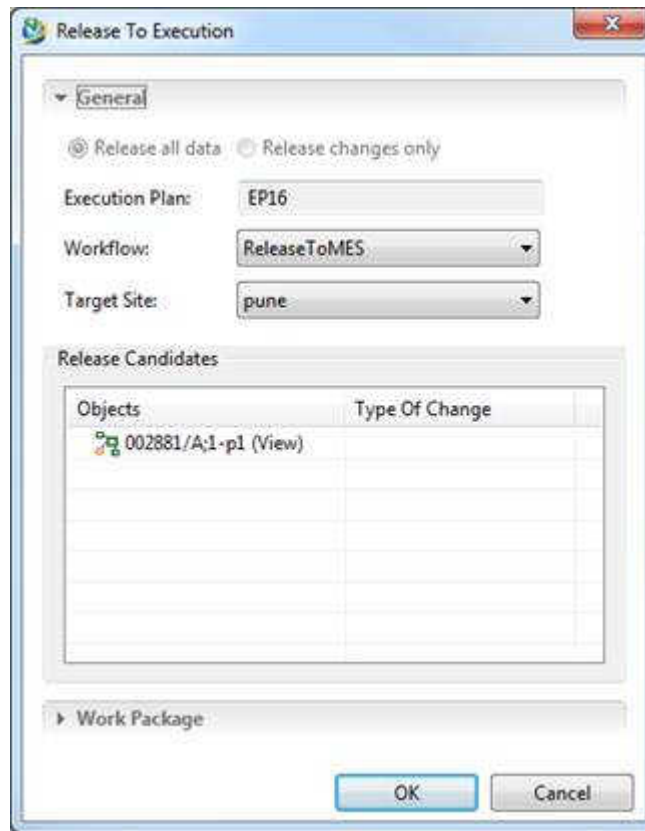
The list of available workflows is stored in the **MEMESWorkflowAndApplicationInterfaceType** preference. The **ReleaseToMES** workflow is delivered with the product.

4. Click the arrow beside the **Target Sites** box to select the sites to which you want to send the data.

The list of available sites is stored in the **MEMESSites** preference.

The line you see in the **Objects** list is the top line for a full release and delta lines are displayed for a delta release.

Teamcenter displays the name of the application interface object as the name of the work package.



5. Click **OK**.

Even if you already validated the selected data, Teamcenter validates the data again as the data may have changed since the validation. It then initiates the workflow and creates the work package. The **Release to Execution** command creates the application interface and the workflow adds an application interface request to the application interface. You can view the status of the AI requests in the application interface viewer.

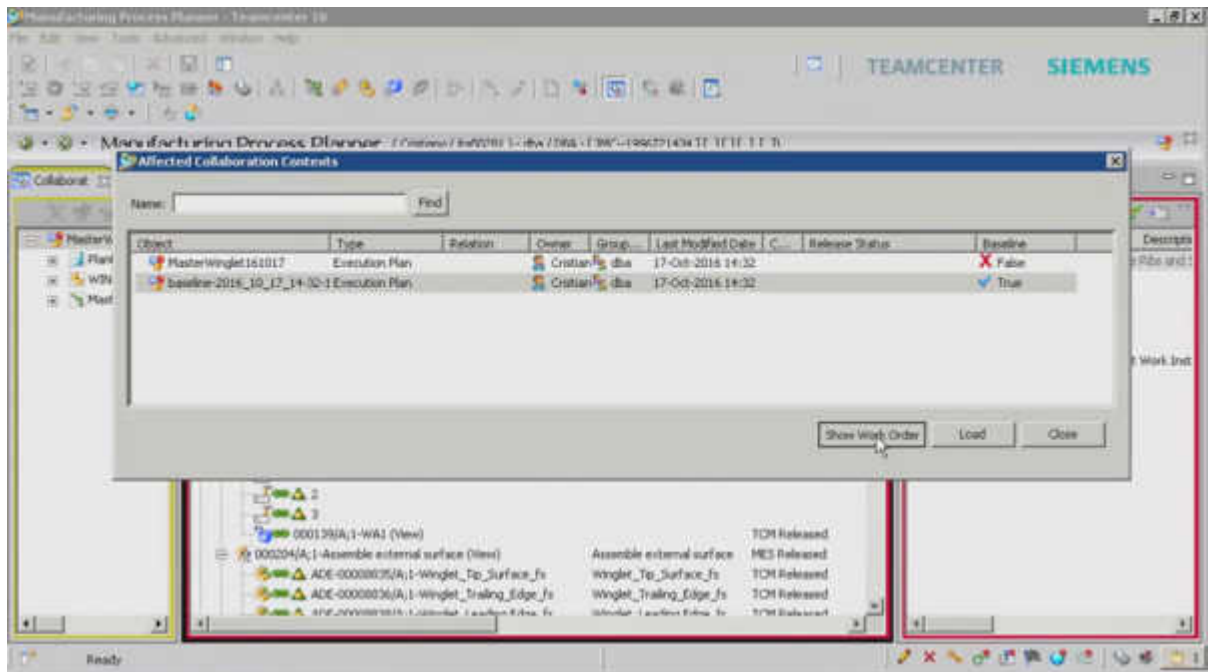
6. (Optional) View the log file to see any errors generated by the workflow. The log file is saved as a dataset of type **TEXT** and added to the workflow as a reference attachment in the task attachments **References** folder. To view this log file:
 - a. Select the execution plan and choose **Send To→ Workflow Viewer**. **Workflow Viewer** displays the completed workflow process.
 - b. Right-click the validation task and choose **Properties**.
 - c. In the **References** box, double-click the log file to open it.

Release an execution plan to the MES (CLM integration)

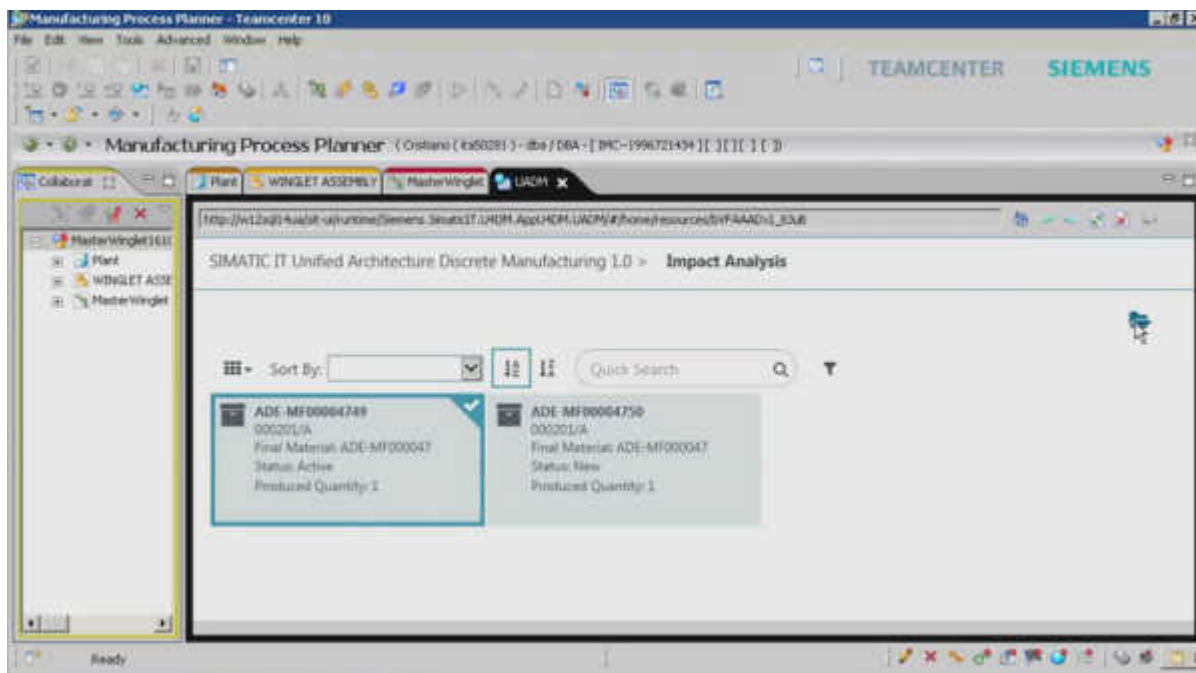
View related work orders to assess the impact of change on the shop floor work order in progress

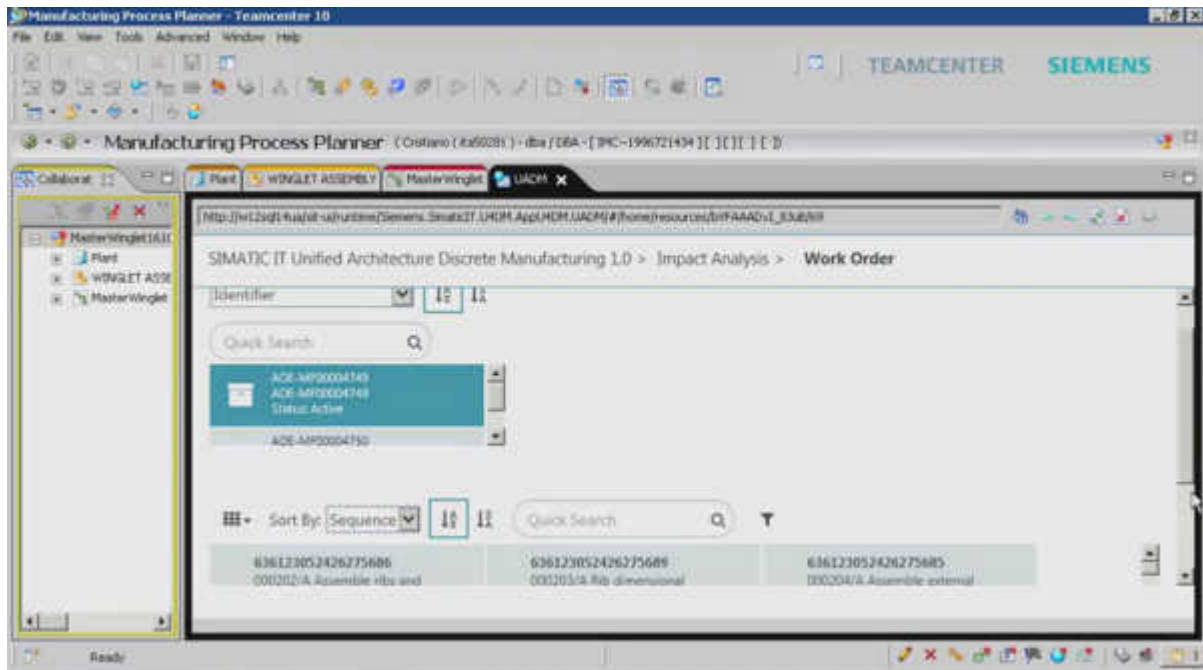
Before you release the changes to the MES Integration, you can view related work orders to assess the impact of change on the shop floor work order in progress.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. Right-click a line in a structure and choose **Tools**→**Find Affected Collaboration Contexts**.



3. Select the appropriate baseline in the **Affected Collaboration Contexts** dialog box and click **Show Work Order**.





Release an execution plan (CLM integration)

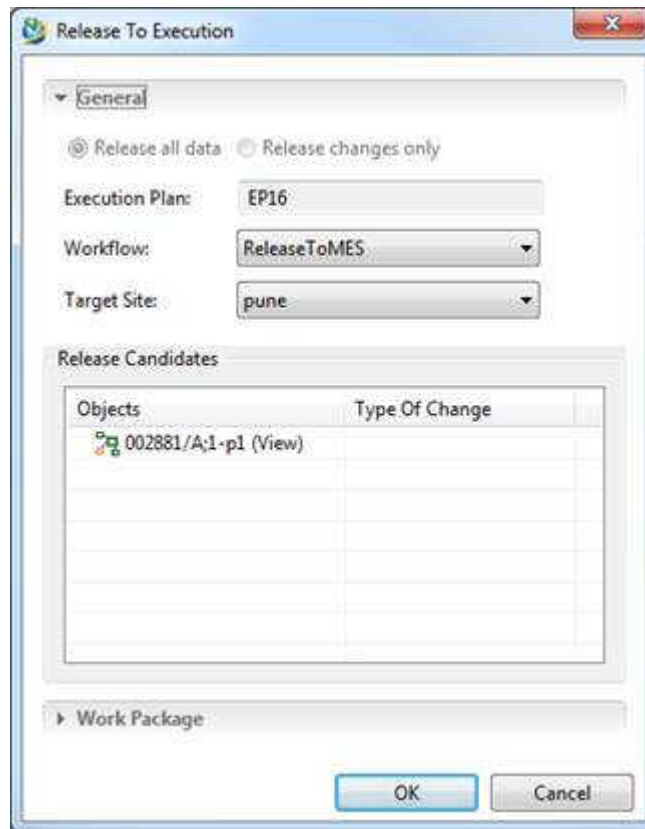
1. Select a valid execution plan from the **Collaboration Context Tree** view in Manufacturing Process Planner or Part Planner.
2. Choose the **Tools→MES→Release to Execution** menu command or choose **Release to Execution** from the shortcut menu.
3. Choose the workflow that you want to use to transfer the data.

The list of available workflows is stored in the **MEMESWorkflowAndApplicationInterfaceType** preference. The **ReleaseToMES** workflow is delivered with the product.

4. Click the arrow beside the **Target Sites** box to select the sites to which you want to send the data.

The list of available sites is stored in the **MEMESSites** preference.

The line you see in the **Objects** list is the top line for a full release and delta lines are displayed for a delta release.



5. Click **OK**.

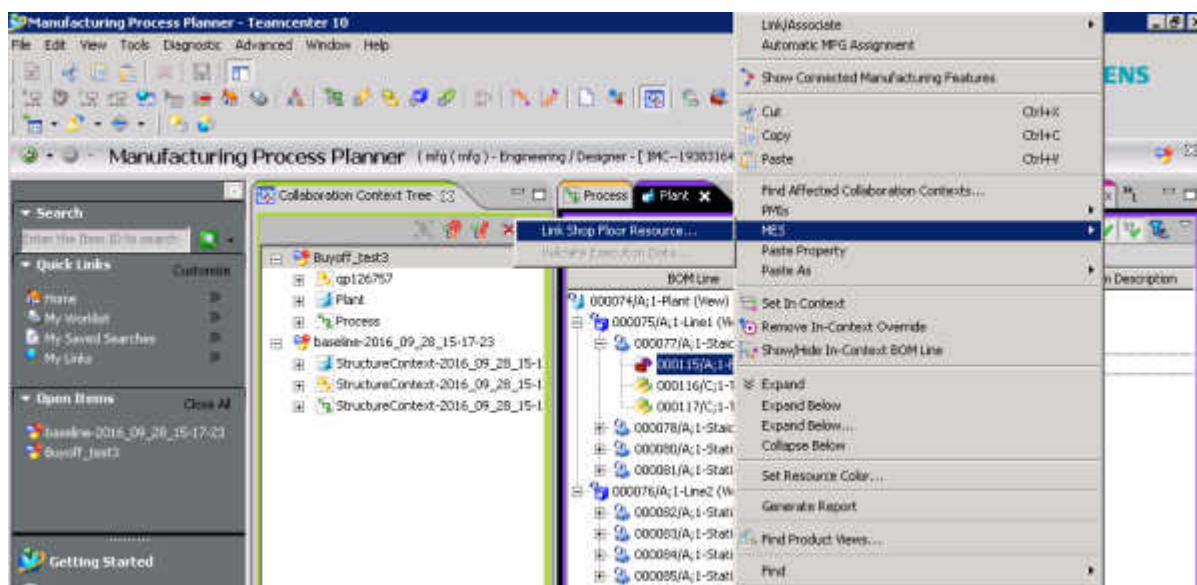
Even if you already validated the selected data, Teamcenter validates the data again as the data may have changed since the validation. It then initiates the workflow and creates a baseline and record object. The **Release to Execution** command exports the BOP in a lightweight JSON format and persists this data as an As Planned BOP in SIMATIC IT.

6. (Optional) **View the log file available on the record object.**

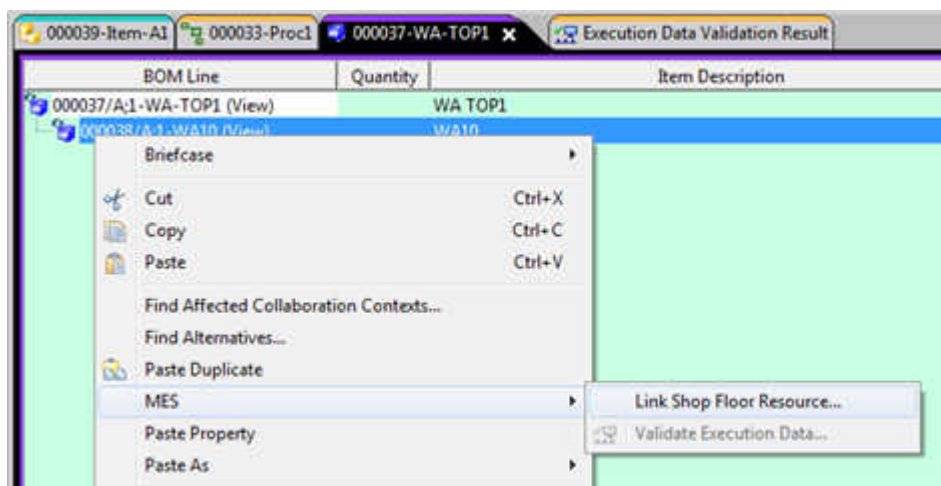
Link the plant structure in Teamcenter to the SIMATIC IT plant structure

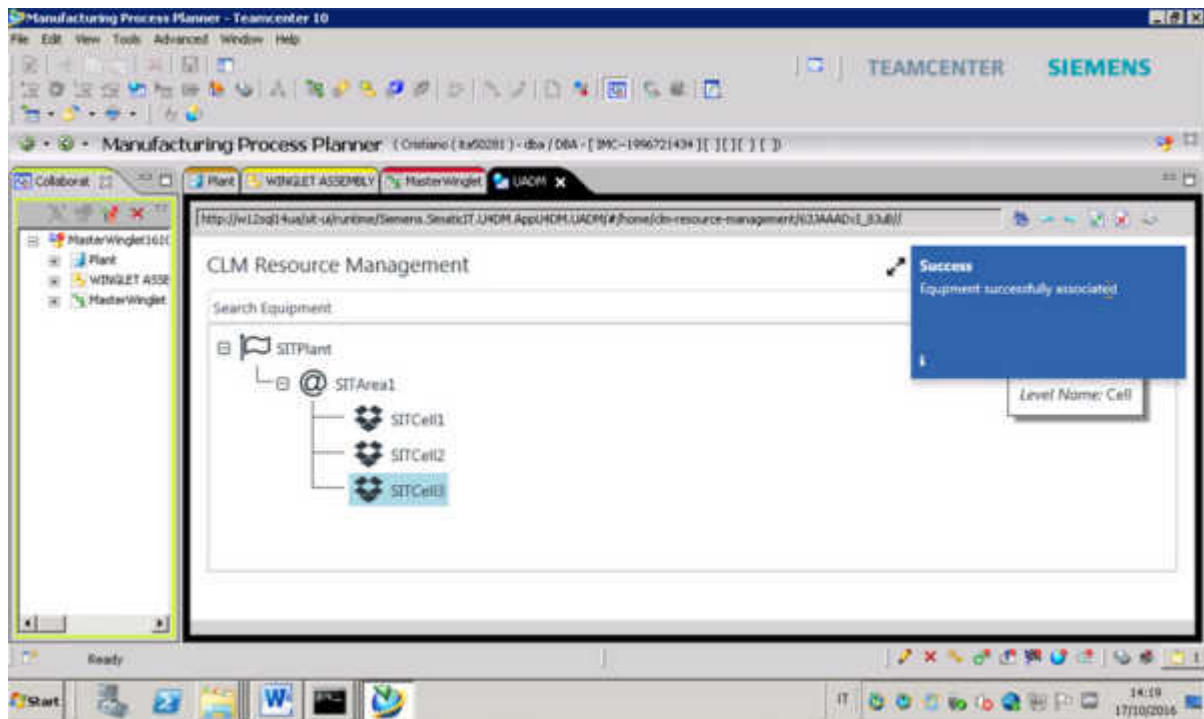
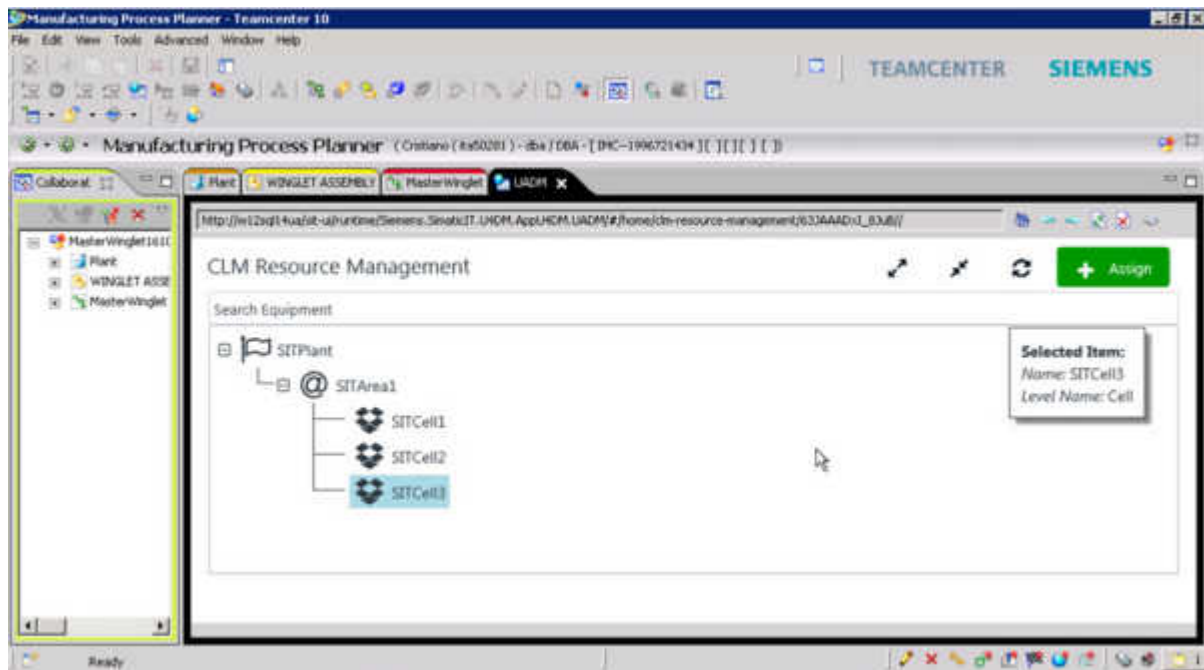
You can link Teamcenter objects such as work area and resources to SIMATIC IT plant model.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. Click the Plant view and select a work area or a resource line (for example, equipment).
3. Right click and choose **MES→Link Shop Floor Resource** to open the SIMATIC IT web page.



4. Select the work area or resource line to which you want to link to in the SIMATIC IT plant structure and click **Assign**.





Release selected BOP lines to the MES

You can send selected BOP lines from the structure to the MES Integration. The system then does not apply any release status to the selected lines, unlike when you release an execution plan to the MES. However, you must perform a full release once before sending the selected BOP lines to the MES.

The system administrator must enable the **MEMESSendWorkflows** site preference and set it to the **SendToMES** default value before you can release selected BOP lines to the MES. In addition, the workflow administrator must set the **-export_as_fai=true** argument to the **MESINTEG_ValidateReleaseAndExport** rule handler. If this property is set to **true**, the work package is considered as a part of the **Send to MES** command.

1. Select a valid execution plan from the **Collaboration Context Tree** view in Manufacturing Process Planner or Part Planner.
2. Select the BOP lines in the structure you want to send to the MES.
3. Choose the **Tools→MES→Send Line(s) to MES** menu command or choose **Send Line(s) to MES** from the shortcut menu.

Note:

You must perform a full release once before sending the selected BOP lines to the MES.

4. Choose the **SendToMES** workflow to transfer the data.

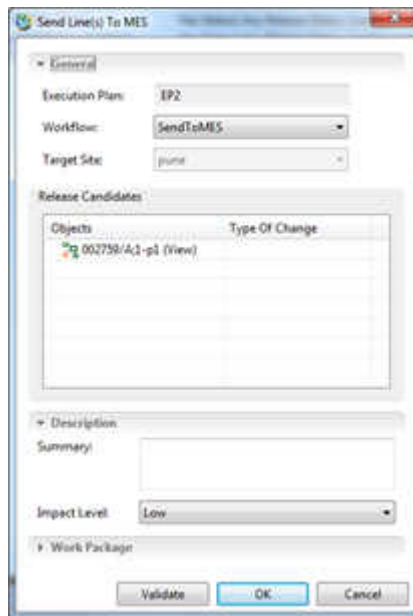
Tip:

The list of available workflows is stored in the **MEMESSendWorkflows** preference.

5. Click the arrow near the **Target Sites** box to select the sites to which you want to send the data.

The list of available sites is stored in the **MEMESSites** preference.

Teamcenter displays the name of the application interface object as the name of the work package.



6. Click **OK**.

Even if you have already validated the selected data, Teamcenter validates the data again as the data may have changed since the validation. It then initiates the workflow and creates the work package. The **Release to Execution** command creates the application interface and the workflow adds an application interface request to the application interface. You can view the status of the AI requests in the application interface viewer.

7. (Optional) View the log file to view any errors generated by the workflow. The log file is saved as a dataset of type **TEXT** and added to the workflow as a reference attachment in the task attachments **References** folder. To view this log file:
 - a. Select the execution plan and choose **Send To → Workflow Viewer**. **Workflow Viewer** displays the completed workflow process.
 - b. Right-click the validation task and choose **Properties**.
 - c. In the **References** box, double-click the log file to open it.

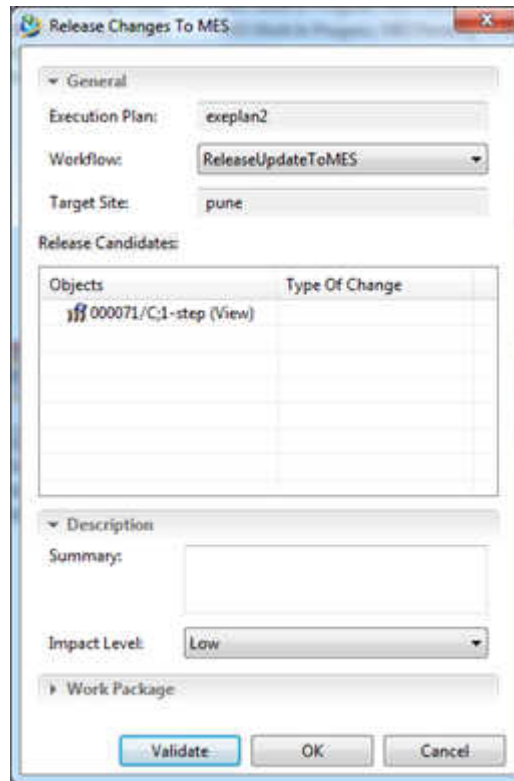
Release changes to the MES

1. Select an execution plan and click .

Teamcenter checks whether the execution plan has already been released to the MES and displays the **Release Options** dialog box.

2. Select **Release changes only** and click **OK**.

Teamcenter displays the **Release changes to MES** dialog box with the name of the execution plan and the name of a workflow.



To track changes for each update to the MES Integration system, the **Release changes to MES** dialog box displays all modifications done to a consumption or work area under an incremental change.

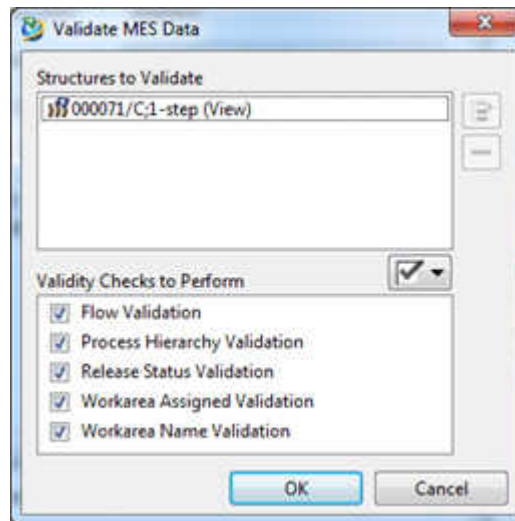
The workflows displayed in this list are those found in the **MEMESUpdateWorkflows** preference in the order that they appear in that preference. In addition, the **Release Candidates** list displays all the lines found in the selected structure that have an **MES Pending To Release** status. The **Type of Change** column displays the nature of the change. If it is the result of an incremental change, you can click the entry in this column to see the change. If the change is the result of revising the line, the cell remains empty.

3. (Optional) Type a description of the work package and choose an impact level.
4. (Optional) Open the **Work Package** section to display the name and creation date of the work package.

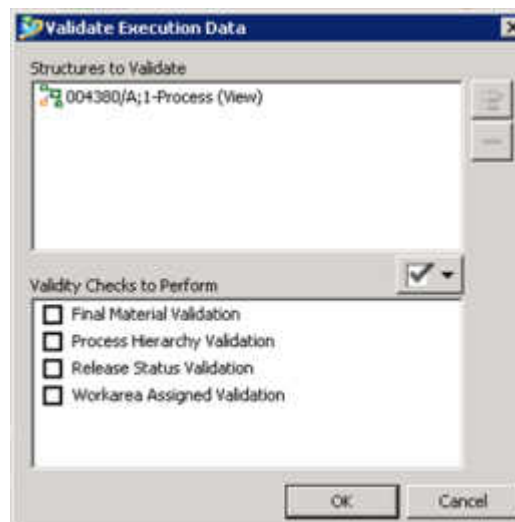
You can use the name to search for the work package in My Teamcenter.

5. To validate only the incremental changes (delta), click the **Validate** button to open the **Validate MES Data** dialog box.

- Using classic integration:



- Using CLM integration:



After releasing an execution plan (classic integration)

View audit logs to understand PLM MES bridge failures

After you release the changes to the MES Integration, you can view audit logs to understand where the PLM MES bridge may have failed. Audit logs display the logged date and time, event type, error messages, and other details. You can optionally use the **audit_purge** utility to archive logs.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. In the **Summary** panel, select the **Audit Logs** tab.

3. Look for error messages and take corrective action as appropriate, for example, the system may display an error message such as *no work area is assigned to a specific object*. In such a case, you can assign a work area to the object and release the change again to the MES Integration.

Creating a new order from a master execution plan

To create a new order from an existing master plan, you must save the process to a new execution plan and assign a new order ID and unit effectivity.

Tip:

Use the **Find Affected Collaboration Contexts** command to help you find all the execution plans related to a particular process.

Load last released execution plan

When Teamcenter releases an execution plan to the MES, it makes a snapshot of the state of the structures at the time of release. This includes all the structures, configuration, and positioning in effect at the time of release. You can load this execution plan and compare it to the current state of the plan, for example, if work has continued on the structure and you want to know what has changed.

- Select the execution plan and click . Teamcenter restores the last released execution plan.

Monitor status of the transfer

After releasing the execution plan to the MES, you can monitor the status of the transfer to see which transactions are complete and which are still in an intermediate state.

Do any of the following to monitor the status of the data transfer:

- Use Workflow Viewer.
When you release an execution plan to the MES, Teamcenter creates a workflow to transfer the data. You can monitor the status of this workflow using Workflow Viewer. You can:
 - See who is assigned to the task.
 - Open the references on the **Attachments** pane to display the log file for the workflow process.

Tip:

You can copy the name of the work package (consisting of the name of the execution plan, the name of the workflow template, and a timestamp) from the **Release to Execution** success message to search for it in the Workflow Viewer. This helps you check the progress of the transfer.

- Monitor the application interface request in My Teamcenter.
You can see the status of the application interface request that is generated by the **Release to Execution** workflow in the **Viewer** view in My Teamcenter.

1. Search for a work package.
2. Select the application interface in the **Search Results** view and choose **Open With→Viewer**.

Teamcenter displays a list of AI requests that are associated with the work package. In addition to the self-explanatory entries such as name and date created, the AI viewer shows the following columns.

Column	Description
State	This value is set by Teamcenter and provides an indication about the request status. The possible values are: • New • Processing • Pending • Communicating • Completed
State Message Status	This value is set by the MES and provides information about the data transfer from the MES side. The possible values are: • Package Retrieved The PLM XML generated by the AI request has arrived at the MES. • Files Retrieved The files are downloaded locally. • In-process The work package is being processed, analyzed and transformed to the MES data model. • Process-Failed There are nonrecoverable errors while transforming the data to the MES mode which means that all of the data has not entered the MES.

Column	Description
	• Success The data is successfully processed and transformed into the MES model.
Status	This value is set by the MES and reflects the severity of the issue. This value is usually Normal and is only set to Aborted if an error occurs.
Status Message	This value is set by the MES. It provides a description of the Status entry.

The table displays some example messages displayed in the viewer. The manufacturing execution system returns one of the following states:

- **Package Retrieved**
The PLM XML generated by the AI request has arrived at the manufacturing execution system.
- **Files Retrieved**
The files are downloaded locally at the manufacturing execution system.
- **Process Failed**
There are nonrecoverable errors in the transformation of the data into the manufacturing execution system model so that the data has not entered the system.
- **Success**
The data is successfully processed and transformed into the manufacturing execution system model.

3. (Optional) Double-click the AI request to see the PLM XML created by the workflow.

Find a work package

If you initiate multiple transfers, you can search for a specific work package using a dedicated query in the **Search** view.

1. Open the **Search** view by clicking ▼ in the navigation pane and choosing **Advanced**.
2. Choose a search type by clicking  in the view toolbar and selecting **More**.
3. From the **System Defined Searches** list, select **MES Work Package** and click **OK**.
4. Enter the execution plan name and any other search criteria you know.

Tip:

The work package name is generally the name of the execution plan that you created.

You must use the following values to query using the **State** attribute.

Value	Interpretation
0	New
1	Processing
2	Pending
3	Communicating
4	Completed

- Click  to execute the search.

Teamcenter displays the search results in the **Search Results** view. The search returns the application interface objects matching the search criteria. You can open an AI object in the viewer to see the application interface request.

Working with shop floor nonconformance issues

During execution, the shop floor operator may discover issues that need to be communicated to engineering. The shop floor operator creates a detailed issue report in the manufacturing execution system and submits it to Teamcenter. Teamcenter creates a shop floor issue business object and initiates a nonconformance workflow. The workflow routes the issue to the concerned planner who then reviews, analyzes, and takes appropriate action on the issue reported. When the planner is finished, Teamcenter marks the review process task as complete and finishes the nonconformance workflow.

Before you can perform issue management, you must have the following:

- The **MES Issue Management** feature installed in TEM.
- A Teamcenter workflow that describes how nonconformance issues are handled.
- The workflow name added to the **Mim0NonConformanceWorkflow** global constant.

The following is a typical nonconformance workflow.

1. The shop floor operator runs the execution based on work instructions sent to the MES.
2. The operator encounters an issue.

3. The operator opens a nonconformance dialog box in the MES and adds information about the problem.
4. The operator adds markups to the work instructions text, 2D images, and 3D models. The markups can consist of basic markups such as a circle around an area or more involved markups such as the following:
 - 2D markup to mark up images
 - 3D markup to mark up product views
 - A combination of 2D and 3D markups
5. (Optional) The operator creates a new drawing and attaches it to the nonconformance issue. For example, the operator captures the work instruction graphics and adds some comments on top of them.
6. (Optional) The operator attaches an execution plan with the **Reference Items** relationship to the issue report.
7. (Optional) The operator creates a Web link object with the **Reference Items** relationship, specifies a URL, and attaches it to the issue report.
8. The MES supervisor decides whether the issue requires planning/design intervention. If yes, the MES supervisor opens a detailed issue report and sends all the relevant data to Teamcenter to report the nonconformance to the planner.
9. The MES supervisor adds a failure code in the MES.
10. Upon receipt of the issue, Teamcenter initiates the workflow specified in the **Mim0NonConformanceWorkflow** business constant.
11. In Teamcenter, the planner finds the issue in the **My Worklist** folder.
If the issue has a product view attached:
 - The structures with which the product view was created are loaded (again, if already loaded) with the configuration they had when the product view was created.
 - Teamcenter opens the graphics view on the structure to which the product view is attached and applies the product view.
12. The planner reviews the accompanying documentation and corrects the problem.
13. The planner sends the changed order or master back to the MES.

After releasing an execution plan (CLM integration)

View audit logs to understand PLM MES bridge failures

After you release the BOP to the MES Integration, you can view audit logs to understand the details of various events in the release process and where the PLM MES bridge may have failed. Audit logs display details such as the logged date and time, event type, and error messages. Optionally, you can use the **audit_purge** utility to archive logs.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. Right-click the top line in the collaboration context and choose **MES→Show Related Baselines**.
3. Select the appropriate baseline in the **Show Related Baselines** dialog box and click **Load**.
4. Copy the **As-Planned Record** value from the **Summary** tab, and then perform a search with this value.
5. From the search results, select the record to display this record.
6. To view logs, in the **Summary** tab, click the **Audit Logs** pane.
7. Look for error messages and take corrective action as appropriate, for example, the system may display an error message such as *no work area is assigned to a specific object*. In such a case, you can assign a work area to the object and release the change again to the MES Integration.

Monitor the transfer status

After releasing the execution plan to the MES, you can monitor the status of the transfer to see the transactions that are complete and the ones that are still in an intermediate state.

1. In My Teamcenter, open the collaboration context containing the execution plan.
2. Right-click the top line in the collaboration context and choose **MES→Show Related Baselines**.
3. Select the appropriate baseline in the **Show Related Baselines** dialog box and click **Load**.
4. Copy the **As-Planned Record** value from the **Summary** tab and then perform a search with this value.
5. From the search results, select the record, and right-click to open the **Properties** dialog box.
6. To view the status, view the **Execution Status** box.

Properties

CLI As-Planned Record

Digital Signature State:

Execution Status:

Group ID: Engineering ▼

Has Tracelink: ☒ True ☐ False

General

Status	Refers to
cli0WorkIn-Progress	Applied WIP status when the baseline is created.
cli0Finished	Applied finished status when the As Planned BOP is created successfully in Simatic IT.
cli0Rollback	Applied rollback status when the As Planned BOP is rolled back successfully in Simatic IT.
cli0Failed	Applied failed status when the As Planned BOP rollback failed, authentication failed, or any other exception.
cli0Discarded	Applied discarded status when a baseline workflow is run without a status and the baseline does not have the cli0Finished status.

Run callbacks to register validation checks

You can register callbacks to perform validation checks. In addition to the default callbacks, you can write your own callbacks to customize validations.

The following are the default callback functions for CLM:

- **Process Flow Validation**
- **UoM check for Process Target**
- **Final Material Validation**
- **UoM Check for Operation Consumption**
- **Get_ChildObjects_From_OccType**
- **Get_ChildObjects_From_Relation**
- **Get_Flow_Property**
- **Get_PMI_DCD_lines_Under_Object**
- **Get_Property_from_PMI_DCD**
- **Filter_by_Type**
- **Get_Dataset_File_Property**
- **Get_Root**

You run the **install_callback** utility to register the callbacks and persist them in the database. For example, run the following argument from the Teamcenter command prompt to register the **Process Flow Validation** callback:

```
install_callback -u=infodba -p=password -g=dba -mode=create -type=MFG_ValidationChecksCallback -library=libCliOclminteg -function=CLI_process_flow_validation -name=Process Flow Validation
```

5. View MES data in the embedded browser

Some planner tasks require knowledge about the actual production. Some examples are:

- Plan optimization
Learning about quality measurements from the shopfloor to help identify problem areas where, for example, tolerances are not achieved or nonconformance statistics are high
- Production status before making a change
Learning about the operation currently being executed (running orders of a specific execution plan)
- Use of resources
Extracting information about resource utilization (for a specific tool or robot in a work location)
- Expected part delivery time
The planner browses the MES system from Teamcenter

You can access any Web-based content from the MES system in the Teamcenter Web Browser.

1. Choose **Tools**→**MES**→**Launch MES Web Page**.
2. Log on to the MES system.

The page is redirected to the URL that is stored in the **MEMESURL** preference.

Your administrator can specify whether the Web Browser opens in a separate window or in an embedded view using the **browserWindowMode** preference.